

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12387528
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Madrigal; Octavio Ponce et al.

Face authentication anti-spoofing using interferometry-based coherence

Abstract

Techniques and apparatuses are described that implement face authentication anti-spoofing using interferometry-based coherence. In particular, a face-authentication system uses ultrasound to distinguish between a real human face and a presentation attack that uses instruments to present a version of a human face. The face-authentication system includes or communicates with an ultrasonic sensor, which can detect a presentation attack and notify the face-authentication system. In general, the ultrasonic sensor uses interferometry to evaluate an amount of coherence (or similarity) between reflections observed by two or more transducers. In this way, the ultrasonic sensor can prevent unauthorized actors from using the presentation attack to gain access to a user's account or information.

Inventors: Madrigal; Octavio Ponce (Mountain View, CA), Heistser; Anton (Munich, DE), Amihood; Patrick M. (Palo Alto, CA)

Applicant: Google LLC (Mountain View, CA)

Family ID: 78695842

Assignee: Google LLC (Mountain View, CA)

Appl. No.: 17/930016

Filed: September 06, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230125564 A1	Apr. 27, 2023

Related U.S. Application Data

continuation parent-doc WO PCT/US2021/056622 20211026 PENDING child-doc US 17930016

Publication Classification

Int. Cl.: G06V40/16 (20220101); G01S15/89 (20060101); G06V10/25 (20220101)

U.S. Cl.:

CPC G06V40/172 (20220101); G01S15/89 (20130101); G06V10/25 (20220101); G06V40/161 (20220101); G06V40/168 (20220101);

Field of Classification Search

CPC: G06V (40/16); G06V (40/161); G06V (40/172); G06V (40/173); G06V (10/25); G06V (40/168); G06V (10/46); G06V (40/13); G06V (40/136); G06V (10/89); G06V (30/186); G06V (30/187); G01S (15/89); G01S (15/8906); G01S (15/8909); G01S (15/8913); G01S (7/288); G01S (13/52)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3522764	12/1969	Biber et al.	N/A	N/A
5729604	12/1997	Van Schyndel	N/A	N/A
6104808	12/1999	Alameh et al.	N/A	N/A
6542436	12/2002	Myllyla	N/A	N/A
7801315	12/2009	Watanabe et al.	N/A	N/A
9665784	12/2016	Derakhshani et al.	N/A	N/A
10108871	12/2017	Derakhshani et al.	N/A	N/A
10262188	12/2018	Agassy et al.	N/A	N/A
10311219	12/2018	Bakish	N/A	N/A
12321431	12/2024	Madrigal et al.	N/A	N/A
2016/0371555	12/2015	Derakhshani	N/A	A61B 5/0077
2019/0065875	12/2018	Sheikh Faridul et al.	N/A	N/A
2019/0138828	12/2018	Hogan	N/A	G06V 40/1365
2019/0258841	12/2018	Rasmussen et al.	N/A	N/A
2020/0066132	12/2019	Kleinbeck	N/A	G08G 5/22
2020/0201968	12/2019	Danielsen	N/A	G01S 15/88
2020/0300970	12/2019	Nguyen	N/A	G01S 13/88
2020/0319301	12/2019	Qiu et al.	N/A	N/A
2022/0222466	12/2021	Hassani et al.	N/A	N/A
2023/0126062	12/2022	Madrigal et al.	N/A	N/A
2023/0129068	12/2022	Madrigal et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3447684	12/2018	EP	N/A
2019089432	12/2018	WO	N/A
2023075751	12/2022	WO	N/A

2023075753	12/2022	WO	N/A
2023075754	12/2022	WO	N/A

OTHER PUBLICATIONS

Chen H, Wang W, Zhang J, Zhang Q. Echoface: Acoustic sensor-based media attack detection for face authentication. IEEE Internet of Things Journal. Dec. 13, 2019;7(3):2152-9. (Year: 2019).

cited by examiner

Park JH. Anomaly detection from the signal of low-cost laser device without the false alarm and the missing. IEEE Sensors Journal. Mar. 26, 2018;18(10):4275-85. (Year: 2018). cited by examiner

“International Search Report and Written Opinion”, Application No. PCT/US2021/056622, Jul. 8, 2022, 14 pages. cited by applicant

“International Search Report and Written Opinion”, Application No. PCT/US2021/056631, Jul. 11, 2022, 14 pages. cited by applicant

“International Search Report and Written Opinion”, Application No. PCT/US2021/056601, Jul. 11, 2022, 15 pages. cited by applicant

Chen, et al., “EchoFace: Acoustic Sensor-Based Media Attack Detection for Face Authentication”, Mar. 2020, pp. 2152-2159. cited by applicant

Lai, et al., “A Smart Spoofing Face Detector by Display Features Analysis”, Jul. 2016, 15 pages. cited by applicant

Ming, et al., “A Survey On Anti-Spoofing Methods For Face Recognition with RGB Cameras of Generic Consumer Devices”, Oct. 8, 2020, 53 pages. cited by applicant

Pinto, et al., “Video-Based Face Spoofing Detection through Visual Rhythm Analysis”, Aug. 2012, 8 pages. cited by applicant

“International Preliminary Report on Patentability”, Application No. PCT/US2021/056622, Apr. 30, 2024, 10 pages. cited by applicant

“International Preliminary Report on Patentability”, Application No. PCT/US2021/056631, Apr. 30, 2024, 10 pages. cited by applicant

“International Preliminary Report on Patentability”, Application No. PCT/US2021/056601, Apr. 30, 2024, 11 pages. cited by applicant

“Non-Final Office Action”, U.S. Appl. No. 17/930,026, filed Nov. 18, 2024, 13 pages. cited by applicant

“Non-Final Office Action”, U.S. Appl. No. 17/929,991, filed Nov. 7, 2024, 9 pages. cited by applicant

“Foreign Office Action”, EP Application No. 21811189.6, Mar. 28, 2025, 4 pages. cited by applicant

“Notice of Allowance”, U.S. Appl. No. 17/929,991, filed Apr. 3, 2025, 8 pages. cited by applicant

Primary Examiner: Mistry; Oneal R

Assistant Examiner: Tran; Duy

Attorney, Agent or Firm: Colby Nipper PLLC

Background/Summary

CROSS-REFERENCE TO A RELATED APPLICATION (1) This application is a continuation application of International Application Serial No. PCT/US2021/056622, filed Oct. 26, 2021, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) Face authentication provides a convenient way for users to unlock their device, add security for accessing accounts, or sign transactions, which enhances the user experience. Some face-authentication systems rely on cameras for face authentication. However, it can be challenging for a camera to distinguish between a user's face and a presentation attack that presents an image of the user's face. As such, challenges arise in preventing unauthorized actors from spoofing a face-authentication system that relies on cameras.

SUMMARY

(2) Techniques and apparatuses are described that implement face authentication anti-spoofing using interferometry-based coherence. In particular, a face-authentication system uses ultrasound to distinguish between a real human face and a presentation attack that uses instruments to present a version of a human face. Example instruments can include a piece of paper with a photograph of a user, a screen displaying a digital image of the user, or a mask that replicates the user's face to some degree. The face-authentication system includes or communicates with an ultrasonic sensor, which can detect a presentation attack and notify the face-authentication system. In general, the ultrasonic sensor uses interferometry to evaluate an amount of coherence (or similarity) between reflections observed by two or more transducers. By using interferometry-based coherence, the ultrasonic sensor can distinguish between the presentation-attack instrument and the human face for face authentication anti-spoofing. Using these techniques, the ultrasonic sensor can prevent unauthorized actors from using the presentation attack to gain access to a user's account or information.

(3) Aspects described below include a method performed by an ultrasonic sensor for face authentication anti-spoofing using coherence. The method includes transmitting an ultrasonic transmit signal and receiving at least two ultrasonic receive signals using at least two transducers of the ultrasonic sensor. The at least two ultrasonic receive signals comprise respective versions of the ultrasonic transmit signal that is reflected by an object. The method also includes generating an interferogram based on the at least two ultrasonic receive signals. The interferogram comprises coherence information and phase information. The method additionally includes identifying a coherence feature based on the coherence information of the interferogram. The coherence feature represents an amount of coherence within a region-of-interest. The method further includes detecting a presentation attack based on the coherence feature. The presentation attack attempts to spoof a face-authentication system and the object is associated with the presentation attack. The method also includes preventing the face-authentication system from authenticating the presentation attack.

(4) Aspects described below include an apparatus with an ultrasonic sensor configured to perform any of the described methods.

(5) Aspects described below also include a computer-readable medium comprising instructions which, when executed by a processor, cause an ultrasonic sensor to perform any of the described methods.

(6) Aspects described below also include a system with means for providing face authentication anti-spoofing using interferometry-based coherence.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Apparatuses for and techniques for face authentication anti-spoofing using interferometry-based coherence are described with reference to the following drawings. The same numbers are used throughout the drawings to reference like features and components:

(2) FIG. 1 illustrates example environments in which face authentication anti-spoofing using

interferometry-based coherence can be implemented;

(3) FIG. 2-1 illustrates an example implementation of a face-authentication system as part of a user device;

(4) FIG. 2-2 illustrates example components of an ultrasonic sensor for face-authentication anti-spoofing;

(5) FIG. 2-3 illustrates example sensors for face-authentication anti-spoofing;

(6) FIG. 3 illustrates an example face-authentication system that performs face-authentication anti-spoofing using ultrasound;

(7) FIG. 4-1 illustrates ultrasonic reflection differences between some presentation-attack instruments and a human face;

(8) FIG. 4-2 illustrates received power differences between some presentation-attack instruments and a human face;

(9) FIG. 5 illustrates example positions of a speaker and microphone of a user device;

(10) FIG. 6 illustrates an example implementation of an ultrasonic sensor for face-authentication anti-spoofing;

(11) FIG. 7 illustrates an example scheme implemented by an ultrasonic sensor for face authentication anti-spoofing;

(12) FIG. 8-1 illustrates an example scheme implemented by an ultrasonic sensor to generate a single-channel feature for face authentication anti-spoofing;

(13) FIG. 8-2 illustrates example range-profile data associated with face authentication anti-spoofing;

(14) FIG. 9-1 illustrates an example scheme implemented by an ultrasonic sensor to generate a multi-channel feature for face authentication anti-spoofing;

(15) FIG. 9-2 illustrates example interferograms associated with face authentication anti-spoofing;

(16) FIG. 10-1 illustrates an example scheme implemented by an ultrasonic sensor for performing co-registration;

(17) FIG. 10-2 illustrates another example scheme implemented by an ultrasonic sensor for performing co-registration;

(18) FIG. 10-3 illustrates an additional example scheme implemented by an ultrasonic sensor for performing co-registration;

(19) FIG. 10-4 illustrates yet another example scheme implemented by an ultrasonic sensor for performing co-registration;

(20) FIG. 11-1 illustrates an example scheme implemented by an ultrasonic sensor to generate another single-channel or multi-channel feature for face authentication anti-spoofing;

(21) FIG. 11-2 illustrates example power distribution histograms associated with face authentication anti-spoofing;

(22) FIG. 11-3 illustrates example constant-false-alarm-rate signal-to-noise ratio histograms associated with face authentication anti-spoofing;

(23) FIG. 12-1 illustrates an example scheme implemented by an ultrasonic sensor to generate yet another single-channel or multi-channel feature for face authentication anti-spoofing;

(24) FIG. 12-2 illustrates a first set of standard-deviation graphs associated with face authentication anti-spoofing;

(25) FIG. 12-3 illustrates a second set of standard-deviation graphs associated with face authentication anti-spoofing;

(26) FIG. 13 illustrates an example method for performing face authentication anti-spoofing using ultrasound;

(27) FIG. 14 illustrates another example method for performing face authentication anti-spoofing using interferometry-based coherence;

(28) FIG. 15 illustrates an additional example method for performing face authentication anti-spoofing using variances; and

(29) FIG. 16 illustrates an example computing system embodying, or in which techniques may be implemented that enable use of, face authentication anti-spoofing using interferometry-based coherence.

DETAILED DESCRIPTION

(30) Overview

(31) Face authentication provides a convenient way for users to unlock their device, add security for accessing accounts, or sign transactions, which enhances the user experience. Some face-authentication systems rely on cameras for face authentication. However, it can be challenging for a camera to distinguish between a user's face and a presentation attack that presents an image of the user's face. As such, challenges arise in preventing unauthorized actors from spoofing a face-authentication system that relies on cameras.

(32) Some techniques distinguish between real face and an image of a face by detecting liveliness (e.g., blinking or facial movements). These techniques, however, can require additional capture time, thereby increasing latency involved with performing face authentication. Furthermore, these techniques may be overcome using a video recording of a human face that includes these movements.

(33) To address these problems, this document describes techniques and devices that implement face authentication anti-spoofing using ultrasound. In particular, a face-authentication system uses ultrasound to distinguish between a real human face and a presentation attack that uses instruments to present a version of a human face. Example instruments can include a piece of paper with a photograph of a user, a screen displaying a digital image of the user, or a mask that replicates the user's face to some degree. The face-authentication system includes or communicates with an ultrasonic sensor, which can detect a presentation attack and notify the face-authentication system. In general, the ultrasonic sensor uses interferometry to evaluate an amount of coherence (or similarity) between reflections observed by two or more transducers. By using interferometry-based coherence, the ultrasonic sensor can distinguish between the presentation-attack instrument and the human face for face authentication anti-spoofing. Using these techniques, the ultrasonic sensor can prevent unauthorized actors from using the presentation attack to gain access to a user's account or information.

(34) Example Environment

(35) FIG. 1 is an illustration of example environments **100-1** to **100-5** in which techniques for face authentication anti-spoofing using interferometry-based coherence may be embodied. In the environments **100-1** to **100-5**, a user device **102** performs face authentication **104**. In environments **100-1** and **100-2**, a user **106** controls the user device **102** and uses face authentication **104** to, for example, access an application or to sign (electronically) a transaction. During face authentication **104**, the user device **102** captures one or more images **108** using a camera system. In this case, the image **108** includes the user **106**'s face. Using facial recognition techniques, the user device **102** recognizes the user's face from the image and authenticates the user **106**.

(36) In some cases, the user **106** may wear an accessory **110** (e.g., a hat, a scarf, a headband, glasses, or jewelry), which can make it more challenging for the user device **102** to recognize the user **106**. In this situation, the user device **102** can use ultrasound to determine an absence of a presentation attack and confirm that a real human face is present. With ultrasound, the user device **102** can successfully perform face authentication **104** during situations in which the user **106** is not wearing an accessory **110**, such as in environment **100-1**, and in situations in which the user **106** chooses to wear an accessory **110**, such as in environment **100-2**.

(37) In other situations, an unauthorized actor **112** may have control of the user device **102**. In this case, the unauthorized actor **112** can use various techniques to attempt to spoof (e.g., trick) the user device **102** into granting the unauthorized actor **112** access to the user **106**'s account or information. Environments **100-3**, **100-4**, and **100-5** provide examples of three different presentation attacks **114-1**, **114-2**, and **114-3**, respectively.

(38) In the environment **100-3**, the unauthorized actor **112** presents a medium **116** with a photograph **118** of the user **106** during face authentication **104**. The medium **116** can be a wood-based medium (e.g., paper, cardboard, or poster board), a plastic-based medium (e.g., an acrylic sheet), a cloth-based medium (e.g., a cotton fabric or a polyester fabric), a sheet of glass, and so forth. In some cases, the medium **116** presents a relatively flat surface in which the photograph is attached, printed, or carved into the medium **116**. In other cases, the medium **116** can be curved or attached to an object with a curved surface. During face authentication **104**, the unauthorized actor **112** orients the medium **116** towards a camera of the user device **102** to cause the user device **102** to capture images **108** of the photograph **118** presented on the medium **116**.

(39) In the environment **100-4**, the unauthorized actor **112** presents a device **120** with a display **122** during face authentication **104**. The device **120** can be a smartphone, a tablet, a wearable device, a television, a curved monitor, or a virtual or augmented reality headset. The display **122** can be a light-emitting diode (LED) display or a liquid-crystal display (LCD). On the display **122**, the device **120** presents a digital picture **124** of the user **106**. During face authentication **104**, the unauthorized actor **112** orients the display **122** of the device **120** towards the camera of the user device **102** to cause the user device **102** to capture images **108** of the digital picture **124** presented on the display **122**.

(40) In the environment **100-5**, the unauthorized actor **112** wears a mask **126**, which replicates one or more features of the user **106**'s face in some way. For example, a color of the mask **126** can be approximately similar to the user **106**'s skin color or the mask **126** can include structural features that represent the user's chin or cheekbone structure. The mask **126** can be a rigid plastic mask or a latex mask. During face authentication **104**, the unauthorized actor **112** wears the mask **126** and faces towards the camera of the user device **102** to cause the user device **102** to capture images **108** of the mask **126**.

(41) The presentation attacks **114-1** to **114-3** shown in environments **100-3** to **100-5** can spoof some authentication systems, which may then grant the unauthorized actor **112** access. However, with the techniques of face-authentication anti-spoofing using ultrasound, the user device **102** detects the presentation attacks **114-1** to **114-3** and denies the unauthorized actor **112** access.

(42) With the described techniques, the user device **102** can distinguish between a human face (e.g., a face of the user **106**) and a representation of a face (e.g., the photograph **118**, the digital picture **124**, or the mask **126**) presented by a presentation attack **114**. In particular, the user device **102** uses ultrasound to detect differences in ultrasonic reflections between the human face and the representation of the face. A human face, for instance, is composed of human tissue, which is not as reflective as other types of material, such as plastic. Also, the human face has curves and angles, which can reduce the amount of ultrasonic energy that is reflected directly back to the user device **102**. In contrast, presentation-attack instruments with relatively flat surfaces or uniformly curved surfaces, such as the medium **116**, the display **122**, or a rigid type of mask **126**, can increase the amount of ultrasonic energy that is reflected back to the user device **102**. By analyzing characteristics of the reflected ultrasonic energy, the user device **102** can determine whether a human face or an instrument used in a presentation attack **114** is presented for face authentication **104**. The user device **102** is further described with respect to FIG. 2-1.

(43) Example Face-Authentication System

(44) FIG. 2-1 illustrates a face-authentication system **202** as part of the user device **102**. The user device **102** is illustrated with various non-limiting example devices including a desktop computer **102-1**, a tablet **102-2**, a laptop **102-3**, a television **102-4**, a computing watch **102-5**, computing glasses **102-6**, a gaming system **102-7**, a microwave **102-8**, and a vehicle **102-9**. Other devices may also be used, including a home-service device, a smart speaker, a smart thermostat, a security camera, a baby monitor, a WiFi® router, a drone, a trackpad, a drawing pad, a netbook, an e-reader, a home-automation and control system, a wall display, a virtual-reality headset, and another home appliance. Note that the user device **102** can be wearable, non-wearable but mobile, or relatively

immobile (e.g., desktops and appliances).

(45) The user device **102** includes one or more computer processors **204** and one or more computer-readable medium **206**, which includes memory media and storage media. Applications and/or an operating system (not shown) embodied as computer-readable instructions on the computer-readable medium **206** can be executed by the computer processor **204** to provide some of the functionalities described herein. The computer-readable medium **206** also includes an application **208** or setting, which activates in response to the face-authentication system **202** authenticating the user **106**. Example applications **208** can include a password-storage application, a banking application, a wallet application, a health application, or any application that provides user privacy.

(46) The user device **102** can also include a network interface **210** for communicating data over wired, wireless, or optical networks. For example, the network interface **210** may communicate data over a local-area-network (LAN), a wireless local-area-network (WLAN), a personal-area-network (PAN), a wide-area-network (WAN), an intranet, the Internet, a peer-to-peer network, point-to-point network, a mesh network, and the like. The user device **102** may also include a display (not shown).

(47) The face-authentication system **202** enables the user **106** to access the application **208**, settings, or other resources of the user device **102** using an image of, or that includes, the user **106**'s face. The face-authentication system **202** includes at least one camera system **212**, at least one face recognizer **214**, and at least one ultrasonic sensor **216**. The face-authentication system **202** can optionally include another sensor **218**. Although shown as part of the face-authentication system **202** in FIG. 2-1, the ultrasonic sensor **216** and/or the sensor **218** can be considered separate entities that can communicate (e.g., provide information) to the face-authentication system **202** for face-authentication anti-spoofing. Sometimes the ultrasonic sensor **216** and/or the sensor **218** operate to support other features of the user device **102** in addition to supporting face-authentication anti-spoofing. Various implementations of the face-authentication system **202** can include a System-on-Chip (SoC), one or more Integrated Circuits (ICs), a processor with embedded processor instructions or configured to access processor instructions stored in memory, hardware with embedded firmware, a printed circuit board with various hardware components, or any combination thereof.

(48) The face-authentication system **202** can be designed to operate under a variety of different environmental conditions. For example, the face-authentication system **202** can support face authentication for distances that are approximately 70 centimeters (cm) or less. The distance represents a distance between the user device **102** and the user **106**. As another example, the face-authentication system **202** can support face authentication for a variety of tilt and pan angles of the user device **102** that provide angular views between approximately -40 and 40 degrees. These angles can include tilt angles between approximately -40 to 20 degrees and pan angles between approximately -20 and 20 degrees. With this range of angular views, the face-authentication system **202** can operate during situations in which the user **106** holds the user device **102** and/or situations in which the user device **102** is on a surface and the user **106** approaches the user device **102**.

(49) Additionally, the face-authentication system **202** can be designed to make a decision regarding face authentication **104** within a predetermined amount of time, such as 100 to 200 milliseconds (ms). This time frame can include a total amount of time it takes to capture the image **108**, use ultrasound to determine whether or not a presentation attack **114** occurred, and perform facial recognition on the image **108**.

(50) The camera system **212** captures one or more images **108** for face authentication **104**. The camera system **212** includes at least one camera, such as a red-green-blue (RGB) camera. The camera system **212** can also include one or more illuminators to provide lighting, especially in dark environments. The illuminator can include an RGB light, such as an LED. In some

implementations, the camera system **212** can be used for other applications, such as for taking selfies, capturing pictures for the application **208**, scanning documents, reading barcodes, and so forth.

(51) The face recognizer **214** performs face recognition to verify that a human face presented in the image **108** corresponds to an authorized user **106** of the user device **102**. The face recognizer **214** can be implemented in software, programmable hardware, or some combination thereof. In some implementations, the face recognizer **214** is implemented using a machine-learned module (e.g., a neural network).

(52) The ultrasonic sensor **216** uses ultrasound to distinguish between a human face and a presentation attack **114**. The ultrasonic sensor **216** is further described with respect to FIG. 2-2. The sensor **218** provides additional information to the ultrasonic sensor **216**, which enhances the ultrasonic sensor **216**'s ability to detect the presentation attack **114**. The sensor **218** is further described with respect to FIG. 2-3.

(53) FIG. 2-2 illustrates example components of the ultrasonic sensor **216**. In the depicted configuration, the ultrasonic sensor **216** includes a communication interface **220** to transmit ultrasonic sensor data to a remote device, though this need not be used when the ultrasonic sensor **216** is integrated within the user device **102**. In general, the ultrasonic sensor data provided by the communication interface **220** is in a format usable by the face-authentication system **202**.

(54) The ultrasonic sensor **216** also includes at least one transducer **222**, which can convert electrical signals into sound waves. The transducer **222** can also detect and convert sound waves into electrical signals. These electrical signals and sound waves may include frequencies in an ultrasonic range.

(55) A frequency spectrum (e.g., range of frequencies) that the transducer **222** uses to generate an ultrasonic signal can include frequencies in the ultrasonic range, which includes frequencies between approximately 20 kilohertz (kHz) and 2 MHz. In some cases, the frequency spectrum can be divided into multiple sub-spectrums that have similar or different bandwidths. For example, different frequency sub-spectrums may include 30 to 500 kHz, 30 to 70 kHz, 80 to 500 kHz, 1 to 2 MHz, 20 to 48 kHz, 20 to 24 kHz, 24 to 48 kHz, 26 to 29 kHz, 31 to 34 kHz, 33 to 36 kHz, or 31 to 38 kHz.

(56) These frequency sub-spectrums can be contiguous or disjoint, and the transmitted signal can be modulated in phase and/or frequency. To achieve coherence, multiple frequency sub-spectrums (contiguous or not) that have a same bandwidth may be used by the transducer **222** to generate multiple ultrasonic signals, which are transmitted simultaneously or separated in time. In some situations, multiple contiguous frequency sub-spectrums may be used to transmit a single ultrasonic signal, thereby enabling the ultrasonic signal to have a wide bandwidth.

(57) For face authentication anti-spoofing, the ultrasonic sensor **216** can use frequencies that provide a particular range resolution that is advantageous for face authentication anti-spoofing. As an example, the ultrasonic sensor **216** can use a bandwidth that can achieve a range resolution of approximately 7 cm or smaller (e.g., approximately 5 cm or approximately 3 cm). An example bandwidth can be at least 7 kHz. The frequencies can also be selected to support a particular detection range for face authentication, such as approximately 70 cm. Example frequencies include frequencies between approximately 31 to 38 kHz.

(58) Sometimes the ultrasonic sensor **216** supports other features in the user device **102** in addition to face authentication anti-spoofing. These other features can include presence detection or hand-grip detection. In this case, the ultrasonic sensor **216** can dynamically switch between different operational configurations that are optimized for the active feature. For example, the ultrasonic sensor **216** can use frequencies between approximately 26 and 29 kHz for presence detection and frequencies between approximately 31 and 38 kHz for face authentication anti-spoofing.

(59) In an example implementation, the transducer **222** of the ultrasonic sensor **216** has a monostatic topology. With this topology, the transducer **222** can convert the electrical signals into

sound waves and convert sound waves into electrical signals (e.g., can transmit or receive ultrasonic signals). Example monostatic transducers may include piezoelectric transducers, capacitive transducers, and micro-machined ultrasonic transducers (MUTs) that use microelectromechanical systems (MEMS) technology.

(60) Alternatively, the transducer **222** can be implemented with a bistatic topology, which includes multiple transducers located at different positions on the user device **102**. In this case, a first transducer converts the electrical signal into sound waves (e.g., transmits ultrasonic signals), and a second transducer converts sound waves into an electrical signal (e.g., receives the ultrasonic signals). An example bistatic topology can be implemented using at least one speaker and at least one microphone of the user device **102**. The speaker and microphone can be dedicated for operations of the ultrasonic sensor **216**. Alternatively, the speaker and microphone can be shared by both the user device **102** and the ultrasonic sensor **216**. Example positions of the speaker and microphone are further described with respect to FIG. 5.

(61) The ultrasonic sensor **216** includes at least one analog circuit **224**, which includes circuitry and logic for conditioning electrical signals in an analog domain. The analog circuit **224** can include a waveform generator, analog-to-digital converters, amplifiers, filters, mixers, phase shifters, and switches for generating and modifying electrical signals. In some implementations, the analog circuit **224** includes other hardware circuitry associated with the speaker or microphone.

(62) The ultrasonic sensor **216** also includes one or more system processors **226** and at least one system medium **228** (e.g., one or more computer-readable storage media). The system processor **226** processes the electrical signals in a digital domain. The system medium **228** includes a spoofing detector **230**. The spoofing detector **230** can be implemented using hardware, software, firmware, or a combination thereof. In this example, the system processor **226** implements the spoofing detector **230**. The spoofing detector **230** processes responses (e.g., electrical signals) from the transducer **222** to detect a presentation attack **114**. The spoofing detector **230** can be implemented, at least in part, using a heuristics module or a machine-learned module (e.g., a neural network).

(63) In some implementations, the ultrasonic sensor **216** uses information provided by the sensor **218**. For example, the sensor **218** provides the ultrasonic sensor **216** information about a location of an object (e.g., the user **106**'s face or a presentation-attack instrument). This information can include range and/or angle measurements. In this way, the ultrasonic sensor **216** can use the location information provided by the sensor **218** to customize (e.g., filter or normalize) ultrasonic data prior to determining whether or not a presentation attack **114** exists. In some cases, this customization is performed before the ultrasonic sensor **216** independently measures the location of the object using ultrasonic techniques. Optionally, the ultrasonic sensor **216** can utilize the location information provided by the sensor **218** to enhance an accuracy of measuring the object's location using ultrasonic techniques.

(64) As another example, the ultrasonic sensor **216** can utilize motion data provided by the sensor **218**. In particular, the ultrasonic sensor **216** can modify the ultrasonic data to compensate for motion identified by the motion data. Example sensors **218** are further described with respect to FIG. 2-3.

(65) FIG. 2-3 illustrates example sensors **218** for face-authentication anti-spoofing. In the depicted configuration, the sensor **218** can include a phase-difference sensor **232**, an RGB sensor **234**, an inertial measurement unit (IMU) **236** (IMU **236**), and/or a radio-frequency-based sensor **238** (RF-based sensor **238**). In some implementations, the phase-difference sensor **232** is part of a front camera **240** of the camera system **212**. The front camera **240** can also be used to capture the image **108** for face authentication. The phase-difference sensor **232** can measure an object's distance based on detected phase differences. With the measured distance, the ultrasonic sensor **216** can determine a region-of-interest associated with the object and perform distance normalization to compensate for the distance between the object and the ultrasonic sensor **216**, which can vary in

different face-authentication situations. In particular, the ultrasonic sensor **216** calibrates the intensity or amplitude information associated with the object based on the measured distance. This enables the ultrasonic sensor **216** to support face authentication anti-spoofing for a variety of different distances.

(66) The RGB sensor **234** can also be part of the front camera **240**. The RGB sensor **234** can measure an angle to an object and provide this angle to the ultrasonic sensor **216**. The ultrasonic sensor **216** can normalize energy or intensity information based on the measured angle. This enables the ultrasonic sensor **216** to better detect backscattering differences between an object used in a presentation attack **114** or a human face and support face authentication anti-spoofing for a variety of different angles.

(67) The inertial measurement unit **236** can measure a motion of the user device **102**. With this motion information, the ultrasonic sensor **216** can perform motion compensation. Additionally or alternatively, the ultrasonic sensor **216** can be activated responsive to the inertial measurement unit **236** indicating that the user device **102** is approximately stationary. In this way, the inertial measurement unit **236** can reduce the quantity of motion artifacts observed by the ultrasonic sensor **216**.

(68) Example RF-based sensors **238** can include an ultra-wideband sensor, a radar sensor, or a WiFi® sensor. Using radio-frequencies, the RF-based sensor **238** can measure a distance and/or angle to the object that is presented for face authentication **104**. With these measurements, the ultrasonic sensor **216** can determine a region-of-interest for detecting the presentation attack and calibrate the intensity or amplitude information. Interactions between the camera system **212**, the face recognizer **214**, the ultrasonic sensor **216**, and the sensor **218** are further described with respect to FIG. 3.

(69) FIG. 3 illustrates an example face-authentication system **202** that performs face-authentication anti-spoofing using ultrasound. In the depicted configuration, the face-authentication system **202** includes the camera system **212**, the face recognizer **214**, and the ultrasonic sensor **216**. The face-authentication system **202** can also optionally include the sensor **218**. The face recognizer **214** is coupled to the camera system **212** and the ultrasonic sensor **216**. The sensor **218** is coupled to the ultrasonic sensor **216**.

(70) During operation, the face-authentication system **202** accepts a request **302** to perform face authentication **104**. In some situations, the request **302** is provided by the application **208** or another service that requires face authentication **104**. In other situations, the request **302** is provided by the sensor **218** responsive to the sensor **218** detecting an interaction with the user **106**. For example, the inertial measurement unit **236** can send the request **302** to the face-authentication system **202** responsive to detecting the user **106** lifting the user device **102**.

(71) In response to receiving the request **302**, the face-authentication system **202** initializes and activates the camera system **212**. The camera system **212** captures at least one image **108** for face authentication **104**. The camera system **212** provides the captured image **108** to the face recognizer **214**.

(72) The ultrasonic sensor **216** performs ultrasonic sensing to detect a presentation attack **114**. This can occur prior to the camera system **212** capturing the image **108**, while the camera system **212** captures the image **108**, or after the camera system **212** captures the image **108**. By analyzing the reflected ultrasonic energy, ultrasonic sensor **216** can determine whether the reflected ultrasonic energy came from an object associated with a presentation attack **114** or a human face. The ultrasonic sensor **216** generates a spoofing indicator **304**, which indicates whether or not the presentation attack **114** is detected. The ultrasonic sensor **216** provides the spoofing indicator **304** to the face recognizer **214**.

(73) The face recognizer **214** accepts the image **108** and the spoofing indicator **304**. If the spoofing indicator **304** indicates that a presentation attack **114** is not detected, the face recognizer **214** performs face recognition to determine whether the user **106**'s face is present within the image **108**.

The face recognizer **214** generates a report **306**, which is provided to the application **208** or the service that sent the request **302**. If the face recognizer **214** recognizes the user **106**'s face and the spoofing indicator **304** indicates absence of the presentation attack **114**, the face recognizer **214** uses the report **306** to indicate successful face authentication. Alternatively, if the face recognizer **214** does not recognize the user **106**'s face and/or the spoofing indicator **304** indicates occurrence of the presentation attack **114**, the face recognizer **214** uses the report **306** to indicate that face authentication failed. This failure indication can cause the application **208** and/or the user device **102** to deny access.

(74) In some implementations, the sensor **218** provides sensor data **308** to the ultrasonic sensor **216**. The ultrasonic sensor **216** uses the sensor data **308** to customize the processing of ultrasonic-based data, which is measured based on the reflected ultrasonic energy. By considering the additional information provided by the sensor **218**, the ultrasonic sensor **216** can enhance its ability to detect the presentation attack **114**.

(75) In some aspects, the ultrasonic sensor **216** can additionally provide information to the face-authentication system **202** or another component of the face-authentication system **202** to support anti-spoofing techniques performed by this entity. For example, the ultrasonic sensor **216** can provide a measured distance of the object to the face-authentication system **202**. With this information, the face-authentication system **202** can analyze the image **108** provided by the camera system **212** and determine whether a size of the object, as presented in the image **108**, comports with a general size of a human face at the measured distance provided by the ultrasonic sensor **216**. In particular, the face-authentication system **202** can determine an upper and lower boundary associated with a size of a human face at the measured distance. If the size of the object in the image **108** is within the boundary, then the face-authentication system **202** does not detect a presentation attack **114**. However, if the size of the object in the image **108** is outside of the boundary, then the face-authentication system **202** detects the presentation attack **114**. In this way, the ultrasonic sensor **216** can also provide ultrasonic-based information that supports anti-spoofing techniques performed by other entities.

(76) Face Authentication Anti-Spoofing

(77) FIG. 4-1 illustrates ultrasonic reflection differences between some presentation-attack instruments and a human face. At **402**, a presentation-attack instrument **404** (e.g., an object associated with the presentation attack **114**) is presented to the user device **102** during face authentication **104**, as shown in environments **100-3** to **100-5**. At **406**, a human face **408** (e.g., the face of the user **106**) is presented to the user device **102** during face authentication **104**, as shown in environments **100-1** and **100-2**.

(78) The presentation-attack instrument **404** can have a different physical structure than the human face **408**. Some presentation-attack instruments **404**, for example, can have a substantially flat or planar surface, such as the medium **116** or the device **120**. Other presentation-attack instruments **404**, such as the mask **126** or other presentation-attack instruments **404** with curved surfaces, can have an approximately piecewise planar surface in which the presentation-attack instrument **404** is composed of a set of substantially flat surfaces. In contrast, the human face **408** can have significantly more pronounced contours and angles relative to the presentation-attack instrument **404**. These contours form the forehead, nose, eye sockets, lips, cheek, chin, and ears of the human face **408**. Based on these structural differences, the ultrasonic-behavior of the presentation-attack instrument **404** can resemble a point target, whereas the ultrasonic-behavior of the human face **408** can resemble a distributed target.

(79) Also, the presentation-attack instrument **404** and the human face **408** can be composed of different materials. These different materials can result in the presentation-attack instrument **404** and the human face **408** having different absorption and reflection properties associated with ultrasound. For example, the human face **408** is composed of human tissue. Some presentation-attack instruments **404** are composed of other materials, such as plastic, which have a higher

reflection coefficient than human tissue.

(80) For both **402** and **406**, the ultrasonic sensor **216** transmits at least one ultrasonic transmit signal **410**. The ultrasonic transmit signal **410** propagates through space and impinges on the presentation-attack instrument **404** and the human face **408**. Due to differences in the physical structures and materials of the presentation-attack instrument **404** and the human face **408**, the ultrasonic transmit signal **410** interacts differently with the presentation-attack instrument **404** and the human face **408**. This interaction is illustrated by reflections **414-1** and **414-2**, which are respectively associated with the presentation-attack instrument **404** and the human face **408**.

(81) Due to the planar-like surface of the presentation-attack instrument **404**, a significant amount of the ultrasonic transmit signal **410** can be reflected back to the user device **102**, as shown by the reflections **414-1** being oriented towards the left in FIG. 4-1. In contrast, the contours of the human face **408** cause the ultrasonic transmit signal **410** to diffuse in many directions, as shown by the reflections **414-2** being oriented in different directions. As such, less ultrasonic energy can be directed back towards the user device **102** from the human face **408** relative to the presentation-attack instrument **404**. Also, if the presentation-attack instrument **404** includes materials with a higher reflection coefficient than human tissue, more ultrasonic energy can be reflected by the presentation-attack instrument **404** relative to the human face **408**. In other words, the reflections **414-1** associated with the presentation-attack instrument **404** can have higher amplitudes than the reflections **414-2** associated with the human face **408**. This difference in amplitude is illustrated by the reflections **414-1** having longer lengths than the reflections **414-2**.

(82) For both **402** and **406**, the ultrasonic sensor **216** receives at least one ultrasonic receive signal **412**. An amplitude of the ultrasonic receive signal **412** is based on the amount of ultrasonic energy that is reflected and the amount of ultrasonic energy that is directed back towards the ultrasonic sensor **216**. In general, the different physical structures and materials of the presentation-attack instrument **404** and the human face **408** can cause the ultrasonic receive signal **412** to have a higher amplitude in **402** than **406**. Based on the characteristics of the ultrasonic receive signal **412**, the ultrasonic sensor **216** can determine whether the presentation-attack instrument **404** or the human face **408** is presented for face authentication **104**. Differences in the ultrasonic-behavior of the presentation-attack instrument **404** and the human face **408** are further described with respect to FIG. 4-2.

(83) FIG. 4-2 illustrates received power differences between the presentation-attack instrument **404** and the human face **408**. A graph **416** has a first dimension associated with received power and a second dimension associated with range (e.g., distance or slant range). In the graph **416**, a dashed line **418** represents the received power associated with the presentation-attack instrument **404**. A solid line **420** represents the received power associated with the human face **408**. In this example, the presentation-attack instrument **404** and the human face **408** are positioned at approximately a same distance from the user device **102**.

(84) As seen in the graph **416**, the presentation-attack instrument **404** produces received power with a higher peak than the human face **408**. This can be due, at least in part, to a higher reflection coefficient of materials that form the presentation-attack instrument **404**. Also, the received power associated with the presentation-attack instrument **404** is concentrated across a smaller range interval whereas the received power associated with the human face **408** is distributed across a larger range interval. This can be due, at least in part, to the point-like nature of the presentation-attack instrument **404** and the distributed nature of the human face **408**. By exploiting these inherent differences, the ultrasonic sensor **216** can employ a variety of techniques to identify whether an object presented for face authentication **104** corresponds to a presentation attack **114** or a human face **408**. These techniques are further described with respect to FIGS. 7 to 15.

(85) FIG. 5 illustrates example positions for a speaker **502** and microphones **504-1** and **504-2** of the user device **102**. Although the example user device **102** of FIG. 5 is shown to include one speakers **502** and two microphones **504-1** to **504-2**, the ultrasonic sensor **216** may operate with one or more

speakers and one or more of microphones at any given time. In situations in which the ultrasonic sensor **216** can use multiple speakers, the ultrasonic sensor **216** can select a speaker that can provide a higher signal-to-noise ratio. The signal-to-noise ratio can be dependent upon performance characteristics of the speaker and/or a position of the speaker relative to the user **106**.

(86) In this implementation, the speaker **502** and the microphones **504-1** and **504-2** are positioned on a same surface of the user device **102**. In this case, this surface also includes a display **506**. The microphone **504-1** is positioned on a different side of the user device **102** than the microphone **504-2**. Consider a plane **508**, which bisects the user device **102** in half. In this case, a first half **510-1** of the user device **102** includes the speaker **502** and the microphone **504-1**. A second half **510-2** of the user device **102** includes the microphone **504-2**.

(87) By placing the microphones **504-1** and **504-2** far apart (e.g., at a distance greater than a wavelength of the ultrasonic transmit signal **410**), the microphones **504-1** and **504-2** can observe different aspects or features of the reflections **414-1** or **414-2**. Using multi-channel techniques, for instance, the ultrasonic sensor **216** can evaluate differences in the responses observed at the microphones **504-1** and **504-2**. Due to the point-like nature of the presentation-attack instrument **404**, the responses observed at the microphones **504-1** and **504-2** can be substantially similar. In contrast, the responses observed at the microphones **504-1** and **504-2** for the human face **408** can be significantly different due to the distributed nature of the human face **408**. Therefore, the ultrasonic sensor **216** can evaluate the differences in the responses to determine whether or not a presentation attack **114** is occurring. An operation of the ultrasonic sensor **216** is further described with FIG. 6.

(88) FIG. 6 illustrates an example implementation of the ultrasonic sensor **216**. In the depicted configuration, the ultrasonic sensor **216** includes the transducer **222**, the analog circuit **224**, and the system processor **226**. The analog circuit **224** is coupled between the transducer **222** and the system processor **226**. The analog circuit **224** includes a transmitter **602** and a receiver **604**. The transmitter **602** includes a waveform generator **606** coupled to the system processor **226**. Although not shown, the transmitter **602** can include one or more transmit channels. The receiver **604** includes multiple receive channels **608-1** to **608-M**, where M represents a positive integer. The receive channels **608-1** to **608-M** are coupled to the system processor **226**.

(89) The transducer **222** is implemented with a bistatic topology, which includes at least one speaker **502** and at least one microphone **504**. In the depicted configuration, the transducer **222** includes multiple speakers **502-1** to **502-S** and multiple microphones **504-1** to **504-M**, where S represents a positive integer. The speakers **502-1** to **502-S** are coupled to the transmitter **602**, and the microphones **504-1** to **504-M** are coupled to respective receive channels **608-1** to **608-M** of the receiver **604**.

(90) Although the ultrasonic sensor **216** in FIG. 6 includes multiple speakers **502** and multiple microphones **504**, other implementations of the ultrasonic sensor **216** can include a single speaker **502** and a single microphone **504**, a single speaker **502** and multiple microphones **504**, multiple speakers **502** and a single microphone **504**, or other types of transducers capable of transmitting and/or receiving. In some implementations, the speakers **502-1** to **502-S** and microphones **504-1** to **504-M** can also operate with audible signals. For example, the user device **102** can play music through the speakers **502-1** to **502-S** and detect the user **106**'s voice using the microphones **504-1** to **504-M**.

(91) During transmission, the transmitter **602** passes electrical signals to the speakers **502-1** to **502-S**, which emit ultrasonic transmit signals **410-1** to **410-S**, respectively. In particular, the waveform generator **606** generates the electrical signals, which can have similar waveforms (e.g., similar amplitudes, phases, and frequencies) or different waveforms (e.g., different amplitudes, phases, and/or frequencies). In some implementations, the system processor **226** passes a configuration signal **610** to the waveform generator **606**. The configuration signal **610** can specify characteristics of the waveform generation (e.g., a center frequency, a bandwidth, and/or a modulation type).

Using the configuration signal **610**, the system processor **226** can customize the ultrasonic transmit signals **410-1** to **410-S** to support an active feature, such as face authentication anti-spoofing or presence detection. Although not explicitly shown, the waveform generator **606** can also communicate the electrical signals to the system processor **226** or the receiver **604** for demodulation. The ultrasonic transmit signals **410-1** to **410-S** may or may not be reflected by an object (e.g., the human face **408** or the presentation-attack instrument **404**).

(92) During reception, the microphones **504-1** to **504-M** receive ultrasonic receive signals **412-1** to **412-M**, respectively. Relative phase differences, frequencies, and amplitudes between the ultrasonic receive signals **412-1** to **412-M** and the ultrasonic transmit signals **410-1** to **410-S** may vary due to the interaction of the ultrasonic transmit signals **410-1** to **410-S** with a nearby object (e.g., the human face **408** or the presentation-attack instrument **404**) or the external environment (e.g., path loss and noise sources). The ultrasonic receive signals **412-1** to **412-M** represent versions of an ultrasonic transmit signal **410** that propagates from one of the speakers **502-1** to **502-S** to one of the microphones **504-1** to **504-M**. The receive channels **608-1** to **608-M** process the ultrasonic receive signals **412-1** to **412-M** and generate baseband receive signals **612-1** to **612-M**.

(93) The system processor **226** includes the spoofing detector **230**. The system processor **226** and/or the spoofing detector **230** can perform functions such as range compression, baseband processing, demodulation, and/or filtering. In general, the spoofing detector **230** accepts the baseband receive signals **612-1** to **612-M** from the receive channels **608-1** and **608-M** and analyzes these signals to generate the spoofing indicator **304**, as further described with respect to FIG. 7.

(94) FIG. 7 illustrates an example scheme implemented by the ultrasonic sensor **216** for face authentication anti-spoofing. In the depicted configuration, the spoofing detector **230** includes a pre-processor **702**, a feature-extractor **704**, and a spoofing predictor **706**. The pre-processor **702** operates on the baseband receive signals **612-1** to **612-M** to provide data in a format that is useable by the feature extractor **704**. The pre-processor **702** can perform filtering, Fourier transforms (e.g., fast-Fourier transforms), range compression, normalization, and/or clutter cancellation. In some implementations, the filtering and/or normalization is based on the sensor data **308**, as further described below.

(95) The feature extractor **704** can perform a variety of functions in order to extract information useful for distinguishing between the presentation-attack instrument **404** and the human face **408**. In some cases, the feature extractor **704** also utilizes location information (e.g., range and/or angle) of the object to extract the appropriate information. The location information can be determined by the ultrasonic sensor **216** using ultrasonic techniques, provided by the sensor **218** via the sensor data **308**, or some combination thereof. Operations of the feature extractor **704** can vary depending on the type of input data the ultrasonic sensor **216** operates on and whether the input data is associated with a single receive channel **608** or multiple receive channels **608-1** to **608-M**. Example feature extractors **704** are further described with respect to FIGS. 8-1, 9-1, 11-1, and 12-1.

(96) The spoofing predictor **706** determines whether or not a presentation attack **114** occurred based on information provided by the feature extractor **704**. The spoofing predictor **706** can include a comparator **708**, a look-up table **710** (LUT **710**), and/or a machine-learned module **712**.

(97) During operation, the pre-processor **702** accepts the baseband receive signals **612-1** to **612-M** and generates complex data **714**. The complex data **714** includes amplitude and phase information (e.g., real and imaginary numbers). Example types of complex data **714** include range-profile data **716** (e.g., data representing an intensity map), an interferogram **718**, range-slow time data **720** (e.g., data representing a range-Doppler map), and/or a power spectrum **722**.

(98) The range-profile data **716** includes amplitude (e.g., intensity) and phase information across a range dimension and a time dimension (e.g., a fast-time dimension). The range dimension can be represented by a set of range bins (e.g., range cells). The time dimension can be represented by a set of time intervals (e.g., chirp intervals or pulse intervals). The interferogram **718** can include coherence and phase information across range and time dimensions. The range-slow time data **720**

includes amplitude and phase information across the range dimension and a Doppler (or slow-time) dimension. The Doppler dimension can be represented by a set of Doppler bins. The power spectra 722 includes power information across a frequency dimension and a time dimension.

(99) In some aspects, the pre-processor 702 can use the sensor data 308 provided by the sensor 218 to calibrate the complex data 714 and/or filter the complex data 714 according to a region-of-interest. As an example, the pre-processor 702 can calibrate amplitude-based information for the range-profile data 716, the interferogram 718, the range-slow time data 720, and/or the power spectra 722 based on a measured location of an object as provided by the sensor data 308. In particular, the pre-processor 702 can normalize the amplitude information based on a measured distance and/or angle to the object, which is presented for face authentication 104.

(100) Additionally or alternatively, the pre-processor 702 can provide a subset of the complex data 714 that is associated with a region-of-interest identified by the distance and/or angle measurement provided by the sensor data 308. The region-of-interest can include the measured location of the object. In some cases, the region-of-interest is centered around the measured location of the object or surrounds at least a portion of the measured location. A size of the region-of-interest can be based on a general size of the human face 408. Additionally or alternatively, the region-of-interest can be based on a region associated with face authentication (e.g., the region-of-interest can include distances up to 70 cm).

(101) The feature extractor 704 extracts one or more features 724 from the complex data 714. Example features 724 can include a single-channel feature 726 and/or a multi-channel feature 728. The feature extractor 704 determines the single-channel feature 726 based on one of the baseband receive signals 612-1 to 612-M. In this way, the single-channel feature 726 is associated with one of the receive channels 608-1 to 608-M. Additionally or alternatively, the feature extractor 704 determines the multi-channel feature 728 using at least two of the baseband receive signals 612-1 to 612-M. The multi-channel feature 728 is associated with two or more of the receive channels 608-1 to 608-M.

(102) In some aspects, the feature extractor 704 can determine the feature 724 that is associated with the object based on a measured location of the object. For example, the feature extractor 704 can identify a region that encompasses the measured location of the object, which can be determined by analyzing the complex data 714 using ultrasonic techniques or based on the sensor data 308 provided by the sensor 218. Within this region, the feature extractor 704 evaluates the complex data 714 to generate the feature 724. In this way, the ultrasonic sensor 216 can ensure that the feature 724 is related to the object that is presented for face authentication and not another object that exists within the environment.

(103) The spoofing predictor 706 analyzes the feature 724 extracted by the feature extractor 704 to detect the presentation attack 114 and generate the spoofing indicator 304. In some examples, the spoofing predictor 706 uses the comparator 708 to compare the feature 724 to a threshold. The threshold is set to enable the ultrasonic sensor 216 to distinguish between the presentation-attack instrument 404 and the human face 408, which may or may not be wearing an accessory 110. In other implementations, the spoofing predictor 706 references a look-up table 710 to determine whether the feature 724 corresponds to the presentation-attack instrument 404 or the human face 408. In yet other implementations, the spoofing predictor 706 uses the machine-learned module 712 to classify the feature 724 as being associated with the presentation-attack instrument 404 or the human face 408.

(104) The spoofing detector 230 generates the spoofing indicator 304, which controls whether the face-authentication system 202 can successfully authenticate the user 106. The spoofing indicator 304 can include one or more elements that indicate whether the object presented for face authentication 104 correlates with a presentation-attack instrument 404 or a human face 408. An example implementation of the spoofing detector 230 is further described with respect to FIG. 8-1.

(105) Single-Channel-Based Face Authentication Anti-Spoofing

(106) FIG. 8-1 illustrates an example scheme implemented by the ultrasonic sensor **216** to generate a single-channel feature **726** for face authentication anti-spoofing. In the depicted configuration, the feature extractor **704** includes a range-profile feature extractor **802**. Optionally, the pre-processor **702** can include a normalizer **804**. The spoofing predictor **706** includes the comparator **708**, the look-up table **710**, and/or the machine-learned module **712**.

(107) During operation, the pre-processor **702** generates the range-profile data **716**. The range-profile data **716** can be generated by performing a Fourier transform on one of the baseband receive signals **612-1** to **612-M**. For implementations that include the normalizer **804**, the pre-processor **702** can normalize the amplitude information within the range-profile data **716** based on the sensor data **308**, which can include a measured location (e.g., a measured distance and/or angle) of an object that is presented for face authentication **104**. Additionally or alternatively, the normalizer **804** can adjust the amplitude information within the range-profile data **716** based on a measured cross-coupling factor or a constant-false-alarm-rate (CFAR) signal-to-noise ratio (SNR). The pre-processor **702** provides normalized range-profile data **806** to the feature extractor **704**. For implementations that do not include the normalizer **804**, the pre-processor **702** can alternatively provide the range-profile data **716** to the feature extractor **704**. Also, these techniques can be similarly applied using range-slow time data **720**.

(108) The range-profile feature extractor **802** generates a single-channel feature **726** based on the range-profile data **716** or the normalized range-profile data **806**. Example single-channel features **726** can include a peak-amplitude feature **808**, an energy-distribution feature **810**, and/or a phase feature **812**. The peak-amplitude feature **808** identifies a peak amplitude within the range-profile data **716** or the normalized range-profile data **806**. The energy-distribution feature **810** identifies a peak energy or a shape of the distributed energy over the range dimension within the range-profile data **716** or the normalized range-profile data **806**. The phase feature **812** identifies phase information within the range-profile data **716** or the normalized range-profile data **806**.

(109) In some aspects, the range-profile feature extractor **802** can determine the single-channel feature **726** that is associated with the object based on a measured location of the object. For example, the range-profile feature extractor **802** can identify a region that encompasses the measured location of the object. Within this region, the range-profile feature extractor **802** evaluates the amplitude and/or phase information to generate the peak-amplitude feature **808**, the energy-distribution feature **810**, and/or the phase feature **812**. In this way, the ultrasonic sensor **216** can ensure that the single-channel feature **726** is related to the object that is presented for face authentication and not another object that exists within the environment.

(110) The spoofing predictor **706** detects whether or not the presentation attack **114** occurred during face authentication **104** and generates the spoofing indicator **304** to communicate this information to the face-authentication system **202**. Consider an example in which the range-profile feature extractor **802** provides the peak-amplitude feature **808** to the spoofing predictor **706**. In this case, the spoofing predictor **706** can use the comparator **708** to compare a peak amplitude identified by the peak-amplitude feature **808** to a threshold **816**. If the peak amplitude is greater than the threshold **816**, the spoofing predictor **706** detects the presentation attack **114**. Alternatively, if the peak amplitude is less than the threshold **816**, the spoofing predictor **706** does not detect the presentation attack **114** (e.g., detects the human face **408**). This example is further described with respect to FIG. 8-2.

(111) FIG. 8-2 illustrates example range-profile data **716** associated with face authentication anti-spoofing. Graphs **818**, **820**, **822**, and **824** illustrate range-profile data **716** for different types of objects, which are presented during face authentication **104**. In particular, the graphs **818**, **820**, **822**, and **824** depict amplitude information for one of the ultrasonic receive signals **412-1** to **412-M** across the range dimension.

(112) The graph **818** represents range-profile data **716** collected when the object is the human face **408**. The graphs **820** and **822** represents range-profile data **716** collected when the object is a piece

of paper and a piece of cardboard, respectively. The piece of paper and the piece of cardboard can include the photograph **118** of the user **106**, as shown in the environment **100-3**. The graph **824** represents range-profile data **716** collected when the object is the device **120** of environment **100-4**. In this case, the device **120** can display the digital picture **124** of the user **106**.

(113) When the object is the human face **408**, the amplitude of the range-profile data **716** is below the threshold **816**, as shown in the graph **818**. However, when the object is the presentation-attack instrument **404**, the amplitude of the range-profile data **716** is above the threshold **816** for at least some range intervals, as shown in graphs **820**, **822**, and **824**. In this way, the ultrasonic sensor **216** can distinguish between the human face **408** and the presentation-attack instrument **404**.

(114) In some implementations, the ultrasonic sensor **216** can identify a range **826** associated with the detected object. The ultrasonic sensor **216** can determine the range **826** through ultrasonic sensing, based on the sensor data **308**, or some combination thereof. At the identified range **826**, the ultrasonic sensor **216** can compare the amplitude of the range-profile data **716** to the threshold **816** to detect the presentation attack **114**.

(115) Similar detection methods can be applied to the other single-channel features **726**. For example, the spoofing predictor **706** can use the comparator **708** to determine whether the extracted energy-distribution feature **810** represents the human face **408** or the presentation-attack instrument **404**. In particular, if the amount of energy at the range **826** is less than a threshold **816**, the ultrasonic sensor **216** does not detect the presentation attack **114**. However, if the amount of energy is greater than the threshold **816**, the ultrasonic sensor **216** detects the presentation attack **114**. Additionally or alternatively, the spoofing predictor **706** can use the look-up table **710** or the machine-learned module **712** to determine whether an overall shape of the energy-distribution feature **810** across the range dimension corresponds to the human face **408** or the presentation-attack instrument **404**. Similar techniques can be used to analyze the phase feature **812** with the look-up table **710** or the machine-learned module **712**.

(116) Although described with respect to a single channel, the operations described in FIG. **8-1** can be performed, in series or in parallel, for more than one, and optionally for each, available receive channel **608-1** to **608-M**. In some implementations, the feature extractor **704** provides multiple features to the spoofing predictor **706**, such as some combination of the peak-amplitude feature **808**, the energy-distribution feature **810**, and/or the phase feature **812**. By analyzing more than one single-channel feature **726**, the spoofing predictor **706** can improve its ability to detect the presentation attack **114**. Another face authentication anti-spoofing technique is further described with respect to FIG. **9-1**.

(117) Multi-Channel-Based Face Authentication Anti-Spoofing Using Interferometry-Based Coherence

(118) FIG. **9-1** illustrates an example scheme implemented by the ultrasonic sensor **216** to generate a multi-channel feature **728** for face authentication anti-spoofing. In this example, the ultrasonic sensor **216** uses interferometry to evaluate an amount of similarity between reflections observed by two or more transducers **222**, such as the microphones **504-1** and **504-2**. Due to the planar-like or uniform structure of the presentation-attack instrument **404**, the ultrasonic receive signals **412-1** to **412-M** produced by the presentation-attack instrument **404** and received by the microphones **504-1** to **504-M** can have relatively similar characteristics. In other words, the amplitudes and phases of the ultrasonic receive signals **412-1** to **412-M** can be relatively similar over a given time interval. The contoured and non-uniform structure of the human face **408**, however, can cause the ultrasonic receive signals **412-1** to **412-M** produced by the human face **408** and received by the microphones **504-1** to **504-M** to be substantially different. In other words, the amplitudes and phases of the ultrasonic receive signals **412-1** to **412-M** can vary significantly over a similar time interval. By using interferometry, the ultrasonic sensor **216** can distinguish between the presentation-attack instrument **404** and the human face **408** for face authentication anti-spoofing.

(119) In the depicted configuration, the pre-processor **702** includes a co-registration module **902**, a

filter **904**, and an interferogram generator **906**. Although not shown, the pre-processor **702** can also include the normalizer **804** of FIG. **8-1**. The feature extractor **704** includes an interferogram feature extractor **908**. The spoofing predictor **706** includes the comparator **708**. Although not shown, other implementations of the spoofing predictor **706** can use the look-up table **710** or the machine-learned module **712**.

(120) Due to the different positions of the microphones **504-1** to **504-M**, an object that is presented for face authentication **104** can be positioned at different ranges within the range-profile data **716** associated with the corresponding receive channels **608-1** to **608-M**. To address this, the co-registration module **902** aligns the range-profile data **716** associated with different receive channels **608-1** to **608-M** across the range dimension. A variety of different types of co-registration modules **902** are further described with respect to FIGS. **10-1** to **10-4**.

(121) The filter **904** filters data provided by the co-registration module **902** based on a region-of-interest, which can be identified based on the sensor data **308**, the range-profile data **716**, or some combination thereof. The region-of-interest can include a spatial region associated with face authentication or a region surrounding a measured location of the object. By filtering the data, the filter **904** can improve the computational speed of later processes and/or remove data that is not associated with the object of interest.

(122) The interferogram generator **906** generates the interferogram **718**. In particular, the interferogram generator **906** performs complex coherence on the range-profile data **716** (or range-slow time data **720**) associated with two of the receive channels **608-1** to **608-M** to generate the interferogram **718**.

(123) During operation, the co-registration module **902** compensates for differences in locations of the microphones **504-1** to **504-M** by generating co-registered range-profile data **910** based on the range-profile data **716** associated with different receive channels **608-1** to **608-M**. Within the co-registered range-profile data **910**, the range associated with the object is similar across the different receive channels **608-1** to **608-M**. The filter **904** filters the co-registered range-profile data **910** to generate filtered co-registered range-profile data **912**.

(124) Prior to the interferogram generator **906**, the range-profile data **716**, the co-registered range-profile data **910**, and the filtered co-registered range-profile data **912** has unique data associated with each of the receive channels **608-1** to **608-M**. The interferogram generator **906** operates on the filtered co-registered range-profile data **912** across pairs of the receive channels **608-1** to **608-M** to generate at least one interferogram **718**. The interferogram **718** includes coherence information and phase information. The coherence information represents an amount of similarity (e.g., coherence or correlation) between the filtered co-registered range-profile data **912** of two of the receive channels **608-1** to **608-M**.

(125) The interferogram feature extractor **908** generates the multi-channel feature **728** based on the interferogram **718**. Example multi-channel features **728** include a coherence feature **914** and a phase feature **916**. The coherence feature **914** represents an amount of coherence over time for a subset of ranges. Sometimes, the coherence feature **914** represents an average coherence.

(126) A coherence value of zero means that the filtered co-registered range-profile data **912** is not correlated across the pair of receive channels **608-1** to **608-M**. In contrast, a coherence value of one means that the filtered co-registered range-profile data **912** is strongly correlated across the pair of receive channels **608-1** to **608-M**. A coherence value between zero and one means that the filtered co-registered range-profile data **912** is partially or weakly correlated across the pair of receive channels **608-1** to **608-M**.

(127) The phase feature **916** represents an amount of variation in the phase information of the interferogram **718** over time for the subset of ranges. Example phase features **916** can represent an average phase difference across a region of the interferogram **718**, a maximum phase difference across the region of the interferogram **718**, and/or a standard deviation across the region of the interferogram **718**.

(128) In some aspects, the interferogram feature extractor **908** can determine the coherence feature **914** and/or the phase feature **916** that is associated with the object based on a measured location of the object. For example, interferogram feature extractor **908** can identify a region that encompasses the measured location of the object. Within this region, the interferogram feature extractor **908** evaluates the filtered co-registered range-profile data **912** to extract the coherence feature **914** and/or the phase feature **916**. In this way, the ultrasonic sensor **216** can ensure that the multi-channel feature **728** is related to the object that is presented for face authentication and not another object that exists within the environment. Additionally or alternatively, the filter **904** can filter the co-registered range-profile data **910** based on the measured location of the object to cause the later-generated multi-channel feature **728** to be associated with the object of interest.

(129) The spoofing predictor **706** uses the comparator **708** to compare the coherence feature **914** and/or the phase feature **916** to a corresponding threshold. For example, the spoofing predictor **706** compares the coherence feature **914** to a coherence threshold **918**. In an example, the coherence threshold **918** is approximately equal to 0.5. If the coherence feature **914** (e.g., the amount of coherence observed for a subset of ranges) is less than the coherence threshold **918** and greater than zero, the spoofing predictor **706** determines that the object presented for face authentication **104** corresponds to the human face **408**. Alternatively, if the coherence feature **914** is greater than the coherence threshold **918** and less than one, the spoofing predictor **706** determines that the object presented for face authentication **104** corresponds to the presentation-attack instrument **404**.

(130) As another example, the spoofing predictor **706** compares the phase feature **916** to the phase threshold **920**. If the phase feature **916** indicates that the phase information within the interferogram **718** is greater than the phase threshold **920**, the spoofing predictor **706** determines that the object presented for face authentication **104** corresponds to the human face **408**.

Alternatively, if the phase feature **916** is less than the phase threshold **920**, the spoofing predictor **706** determines that the object presented for face authentication **104** corresponds to the presentation-attack instrument **404**.

(131) Some implementations of the spoofing predictor **706** can analyze both the coherence feature **914** and the phase feature **916** to determine whether the presentation attack **114** occurred. The spoofing predictor **706** can additionally or alternatively use other techniques, such as those that employ the look-up table **710** or the machine-learned module **712** to analyze the interferogram **718** and determine whether the presentation attack **114** occurred.

(132) The operations described in FIG. 9-1 can be performed for range-profile data **716** provided by a pair of receive channels **608-1** to **608-M**. If the ultrasonic sensor **216** uses more than two receive channels **608-1** to **608-M**, these operations can be performed in series or parallel for multiple pairs of the receive channels **608-1** to **608-M**. Consider an example in which the ultrasonic sensor **216** uses four receive channels (e.g., M equals four). In this case, the pre-processor **702** can generate up to six interferograms **718**, which can be processed by the feature extractor **704** and the spoofing predictor **706** to detect the presentation attack **114**. If multiple interferograms **718** are available, the spoofing predictor **706** can detect the presentation attack **114** responsive to the multi-channel feature **728** associated with one of the multiple interferograms **718** indicating the presence of the presentation-attack instrument **404**. To decrease a probability of the ultrasonic sensor **216** incorrectly determining that a human face **408** presented for face authentication **104** represents the presentation-attack instrument **404**, the spoofing predictor **706** can detect the presentation attack **114** responsive to a predetermined quantity of the interferograms **718** indicating the presence of the presentation-attack instrument **404** (e.g., such as three or more of the multiple interferograms **718**). Example phase information and coherence information provided by the interferogram **718** is further described with respect to FIG. 9-2.

(133) FIG. 9-2 illustrates example interferograms **928** to **932** associated with face authentication anti-spoofing. The interferogram **928** represents data collected when the object is the human face **408**. The interferograms **930** and **932** represent data collected when the object is the device **120** or a

piece of paper, respectively. The piece of paper can include the photograph **118** of the user **106**, as shown in the environment **100-3**.

(134) Each of the interferograms **928** to **932** include phase information **934** and coherence information **936**. Different shades of the phase information **934** represents different phase angles. Different shades of the coherence information **936** represent different coherence amounts. Lighter shades represent higher amounts of coherence, and darker shades represent lower amounts of coherence. Cells of the interferograms **928** to **932** associated with a region-of-interest **938** are identified in FIG. **9-2**. The region-of-interest **938** is based on a measured distance between the object presented for face authentication **104** and the ultrasonic sensor **216**.

(135) Within the region-of-interest **938** of the interferogram **928**, the phase information **934** includes significantly different phases. The coherence information **936** includes many cells that are weakly-correlated with coherence amounts that are less than 0.5. The coherence within the region-of-interest **938** can appear to be random. These characteristics can be due, at least in part, to the contoured and distributed nature of the human face **408**.

(136) Within the region-of-interest **938** of the interferograms **930** and **932**, the phase information **934** depicts similar phases over time. The coherence information **936** within the region-of-interest **938** includes many cells that are strongly-correlated with coherence amounts that are greater than 0.5. These characteristics can be due, at least in part, to the planar and point-like nature of these presentation-attack instruments **404**.

(137) Comparing the region-of-interests **938** between interferograms **928** to **932**, the phase information **934** within the interferograms **930** and **932** has less variation than the phase information **934** within the interferogram **928**. Also, the coherence information **936** within the interferograms **930** and **932** has higher coherence amounts than the coherence information **936** within the interferogram **928**. Additionally, there is less variation between the coherence information **936** within the interferograms **930** and **932** as compared to the coherence information **936** within the interferogram **928**. In general, the multi-channel feature **728** extracted by the feature extractor **704** and the analysis performed by the spoofing predictor **706** enable the ultrasonic sensor **216** to detect these characteristics and appropriately identify an object as corresponding to the human face **408** or the presentation-attack instrument **404**.

(138) The interferometry techniques can also be applied to distinguish between the user **106** wearing an accessory **110** and the presentation-attack instrument **404**. In some examples, the accessory **110** causes the coherence information **936** to have a few strongly-correlated cells. However, these strongly-correlated cells can exist across a broader region in range in comparison to the strongly-correlated cells of the presentation-attack instrument **404**. By analyzing the phase information **934** and/or the coherence information **936** within the interferogram **928**, the ultrasonic sensor **216** can recognize characteristics that distinguish the human face **408** from the presentation-attack instrument **404**.

(139) FIG. **10-1** illustrates an example scheme implemented by the ultrasonic sensor **216** for performing co-registration. In this case, the co-registration module **902** uses low-pass filtering to align the range-profile data **716** associated with the receive channels **608-1** and **608-2**. The co-registration module **902** includes two low-pass filters **1002-1** and **1002-2**. During operation, the low-pass filters **1002-1** and **1002-2** respectively filter the range-profile data **716** of the receive channels **608-1** and **608-2** across the range dimension to produce the co-registered range-profile data **910**. By using the low-pass filters **1002-1** and **1002-2**, the amplitude and/or phase information across adjacent range bins are combined to form a larger composite range bin. Although an object's range bin may differ between the range-profile data **716** of the receive channels **608-1** and **608-2**, the object's range bin can be the same within the larger composite range bins of the co-registered range-profile data **910**. In this way, the co-registration module **902** aligns the range-profile data **716** of the receive channels **608-1** and **608-2** in range.

(140) Although implementing the co-registration module **902** with the low-pass filters **1002-1** and

1002-2 can be relatively simple, the resulting co-registered range-profile data **910** has a lower range resolution relative to the range-profile data **716**. The low-pass filtering also provides localized co-registration. This means that the range bins associated with the object are substantially aligned while other range bins within the range-profile data **716** may not be aligned. To address the reduced range resolution, the co-registration module **902** can instead utilize techniques described with respect to FIG. **10-2**.

(141) FIG. **10-2** illustrates another example scheme implemented by the ultrasonic sensor **216** for performing co-registration. In this case, the co-registration module **902** evaluates maximum returns to align the range-profile data **716** associated with the receive channels **608-1** and **608-2**. The co-registration module **902** includes two peak detectors **1004-1** and **1004-2**, a comparator **1006**, and a shifter **1008**.

(142) During operation, the peak detectors **1004-1** and **1004-2** respectively evaluate the range-profile data **716** associated with the receive channels **608-1** and **608-2** and identify range bins **1010-1** and **1010-2** associated with a peak amplitude. The comparator **1006** compares the range bins **1010-1** and **1010-2** to determine an amount that the range bin **1010-2** differs from the range bin **1010-1**. This difference represents an offset **1012** (e.g., a range offset) between the range-profile data **716** of the receive channel **608-1** and the range-profile data **716** of the receive channel **608-2**. The shifter **1008** shifts the range-profile data **716** of the receive channel **608-2** across the range dimension by an amount identified by the offset **1012**. This results in the shifter **1008** producing shifted range-profile data **1014** of the receive channel **608-2**. The range-profile data **716** of the receive channel **608-1** and the shifted range-profile data **1014** of the receive channel **608-2** are provided as the co-registered range-profile data **910**.

(143) Although slightly more complex than the co-registration module **902** of FIG. **10-1**, the co-registration module **902** of FIG. **10-2** preserves the range resolution provided by the range-profile data **716**. However, this technique still has the localized co-registration problem described above with respect to FIG. **10-1**. To address the localized co-registration, the co-registration module **902** can instead utilize techniques described with respect to FIG. **10-3**.

(144) FIG. **10-3** illustrates an additional example scheme implemented by the ultrasonic sensor **216** for performing co-registration. In this case, the co-registration module **902** evaluates maximum returns to align the range-profile data **716** associated with the receive channels **608-1** and **608-2** and uses interpolation to provide global co-registration. The co-registration module **902** includes the two peak detectors **1004-1** and **1004-2**, the comparator **1006**, the shifter **1008**, a sub-bin range offset detector **1016** (e.g., a sub-pixel range offset detector), and an interpolator **1018**.

(145) During operation, the peak detectors **1004-1** and **1004-2**, the comparator **1006**, and the shifter **1008** generate the shifted range-profile data **1014** of receive channel **608-2**, as described above with respect to FIG. **10-2**. The sub-bin range offset detector **1016** compares the range-profile data **716** of the receive channel **608-1** to the shifted range-profile data **1014** of the receive channel **608-2** to identify another offset **1020**, which is smaller than a size of the range bins. The interpolator **1018** interpolates the shifted range-profile data **1014** based on the offset **1020** to generate interpolated and shifted range-profile data **1022** of the receive channel **608-2**. The range-profile data **716** of the receive channel **608-1** and the interpolated and shifted range-profile data **1022** of the receive channel **608-2** are provided as the co-registered range-profile data **910**.

(146) By using interpolation, the co-registration module **902** of FIG. **10-3** can more accurately align the range-profile data **716** of the receive channels **608-1** and **608-2** relative to the co-registration modules **902** of FIGS. **10-1** and **10-2**. However, this technique also increases the computational burden. Another accurate technique for co-registration is further described with respect to FIG. **10-4**.

(147) FIG. **10-4** illustrates yet another example scheme implemented by the ultrasonic sensor **216** for performing co-registration. In this case, the co-registration module **902** projects the range-profile data **716** to a common grid. The co-registration module **902** includes a common-grid

projector **1024** and an interpolator **1026**.

(148) During operation, the common-grid projector **1024** accepts Euler angles **1028** and geometry information **1030**. The Euler angles **1028** represent an orientation of the user device **102**. The geometry information **1030** provides information about the relative position of the microphones **504-1** and **504-2** associated with the receive channels **608-1** and **608-2**. The geometry information **1030** can specify distances and angles to the microphones **504-1** and **504-2** based on a reference point associated with the user device **102**. By analyzing the Euler angles **1028** and the geometry information **1030**, the common-grid projector **1024** can determine propagation path differences between the object and the microphones **504-1** and **504-2**. In particular, the common-grid projector **1024** can determine the amplitude and/or phase differences that occur due to the different positions of the microphones **504-1** and **504-2** within the user device **102**.

(149) The interpolator **1026** uses information provided by the common-grid projector **1024** to re-sample or interpolate the range-profile data **716** of the receive channel **608-2**. This resampling adjusts characteristics (e.g., amplitude and phase) of the range-profile data **716** to correspond with the common-grid. In some implementations, the common-grid is based on a position of the microphone **504-1**. In this case, the range-profile data **716** of the receive channel **608-1** does not need to pass through the interpolator **1026** and can be provided as part of the co-registered range-profile data **910**. The interpolator **1026** generates projected range-profile data **1032** of the receive channel **608-2**, which can be provided as another part of the co-registered range-profile data **910**.

(150) Face Authentication Anti-Spoofing Using Skewness

(151) FIG. **11-1** illustrates an example scheme implemented by the ultrasonic sensor **216** to generate another single-channel or multi-channel feature for face authentication anti-spoofing. In this example, the ultrasonic sensor **216** evaluates skewness (e.g., asymmetry or adjusted Fisher-Pearson standardized moment coefficient) of an intensity-based probability distribution for at least one receive channel **608**. The different physical structures of the presentation-attack instrument **404** and the human face **408** cause the intensity-based probability distributions associated with these objects to have different shapes. In this manner, the ultrasonic sensor **216** can distinguish between the presentation-attack instrument **404** and the human face **408** for face authentication anti-spoofing.

(152) In the depicted configuration, the pre-processor **702** includes an intensity-based histogram generator **1102**. The feature extractor **704** includes a skewness feature extractor **1104**. The spoofing predictor **706** includes the comparator **708**, the look-up table **710**, and/or the machine-learned module **712**.

(153) During operation, the intensity-based histogram generator **1102** generates at least one histogram **1106** based on the range-profile data **716**. The intensity-based histogram generator **1102** can generate multiple histograms **1106** if the ultrasonic sensor **216** utilizes multiple receive channels **608-1** to **608-M**. Example histograms **1106** include a power distribution histogram **1108** and a constant-false-alarm-rate signal-to-noise ratio histogram **1110**. The power distribution histogram **1108** represents statistical frequencies of different levels of received power. The constant-false-alarm-rate signal-to-noise ratio histogram **1110** (CFAR SNR histogram **1110**) represents statistical frequencies of different constant-false-alarm-rate signal-to-noise ratios of one of the ultrasonic receive signals **412-1** to **412-M**, which has been normalized based on a CFAR kernel.

(154) In some aspects, the intensity-based histogram generator **1102** can filter the range-profile data **716** based on the measured location of the object prior to generating the histogram **1106**. In this way, the intensity-based histogram generator **1102** can cause a later-determined single-channel feature **726** or multi-channel feature **728** to be associated with the object of interest.

(155) The skewness feature extractor **1104** generates the skewness feature **1112** based on the histogram **1106**. The skewness feature **1112** represents the amount of asymmetry or the adjusted Fisher-Pearson standardized moment coefficient. The spoofing predictor **706** analyzes the skewness

feature **1112** to determine whether the object presented for face authentication **104** corresponds to the human face **408** or the presentation-attack instrument **404**. In particular, the spoofing predictor **706** can determine that the object is the presentation-attack instrument **404** if the skewness feature **1112** indicates that the histogram **1106** is positively skewed relative to a threshold.

(156) If multiple histograms **1106** are available, the spoofing predictor **706** can also evaluate differences between the skewness features **1112** of the multiple histograms **1106**. Sometimes skewness differences greater than 0.1 dB can indicate that the object is the presentation-attack instrument **404**. Additionally or alternatively, the spoofing predictor **706** can identify, using the look-up table **710** or the machine-learned module **712**, the presence of heavy tails within one or more of the histograms **1106**. The term “heavy tail” represents a higher statistical frequency occurring towards one end of a distribution relative to a normal distribution. The presence of heavy tails can indicate the presence of the presentation-attack instrument **404**, as further described below. Example power distribution histograms **1108** and constant-false-alarm-rate signal-to-noise ratio histograms **1110** are depicted in FIGS. **11-2** and **11-3**.

(157) FIG. **11-2** illustrates example power distribution histograms **1114** to **1118** associated with face authentication anti-spoofing. The power distribution histogram **1114** represents data collected when the object is the human face **408**. The power distribution histograms **1116** and **1118** represent data collected when the object is a latex mask and a rigid plastic mask, respectively. Each of the power distribution histograms **1114** to **1118** include a first power distribution **1120** associated with the receive channel **608-1** (e.g., associated with the microphone **504-1** of FIG. **5**), and a second power distribution **1122** associated with the receive channel **608-2** (e.g., associated with the microphone **504-2** of FIG. **5**).

(158) The power distributions **1120** and **1122** within the power distribution histogram **1114** have relatively normal distributions. In this case, the skewness of the power distribution **1120** can be approximately -0.7 dB, and the skewness of the power distribution **1122** can be approximately -0.6 dB. Also, a difference between the skewness of the power distributions **1120** and **1122** can be relatively small (e.g., approximately 0.1 dB or less).

(159) In contrast, at least one of the power distributions **1120** and **1122** within the power distribution histograms **1114** and **1118** are skewed in the positive direction relative to the power distributions **1120** and **1122** within the power distribution histogram **1114**. Example skewness of the power distributions **1120** and **1122** within the power distribution histogram **1116** can be approximately -0.2 and -0.9 dB, respectively. In this case, the skewness of the power distribution **1120** is 0.5 dB greater for the latex mask relative to the human face **408**. The power distribution **1122** within the power distribution histogram **1114** also has a non-Gaussian distribution or shape. In particular, the power distribution **1122** has a heavy tail on the right side.

(160) Example skewness of the power distributions **1120** and **1122** within the power distribution histogram **1118** can be approximately -0.7 and -0.7 dB, respectively. In this case, the skewness of the power distribution **1122** is 0.1 dB greater for the rigid plastic mask relative to the human face **408**. In this case, the ultrasonic sensor **216** may utilize other techniques, such as interferometry (of FIG. **9-1**) or other single-channel techniques (of FIG. **8-1**) to distinguish between the human face **408** and the rigid plastic mask.

(161) For some types of presentation-attack instruments **404**, a difference between the skewness of the power distributions **1120** and **1122** can be substantially large (e.g., greater than 0.3 dB), such as in the power distribution histogram **1116** for the latex mask. This is another characteristic that the spoofing predictor **706** can use to distinguish between the human face **408** and the presentation-attack instrument **404**.

(162) This technique can also be applied to other types of presentation-attack instruments **404**, such as the device **120** and a paper-based medium **116**. For both of these presentation-attack instruments **404**, the positive skewness can be even more pronounced. For example, the skewness of the power distributions **1120** and **1122** can be approximately 0.4 and 0.3 dB, respectively, if the object is the

device **120**. As another example, the skewness of the power distributions **1120** and **1122** can be approximately 0.1 and -0.2 dB, respectively, if the object is the paper-based medium **116**.

(163) FIG. **11-3** illustrates example constant-false-alarm-rate signal-to-noise ratio histograms **1124** and **1126** associated with face authentication anti-spoofing. The constant-false-alarm-rate signal-to-noise ratio histogram **1124** represents data collected when the object is the human face **408**. The constant-false-alarm-rate signal-to-noise ratio histogram **1126** represent data collected when the object is the device **120**. Each of the constant-false-alarm-rate signal-to-noise ratio histograms **1124** and **1126** include a first distribution **1128** associated with the receive channel **608-1** (e.g., associated with the microphone **504-1**), and a second distribution **1130** associated with the receive channel **608-2** (e.g., associated with the microphone **504-2**).

(164) The distributions **1128** and **1130** within the constant-false-alarm-rate signal-to-noise ratio histogram **1124** have relatively normal distributions (e.g., Gaussian distributions). In contrast, both distributions **1128** and **1130** within the constant-false-alarm-rate signal-to-noise ratio histogram **1126** have tails on the right side. The tails in the constant-false-alarm-rate signal-to-noise ratio histogram **1126** are heavier (e.g., have a higher statistical frequency) relative to the tails in the constant-false-alarm-rate signal-to-noise ratio histogram **1124**.

(165) Face Authentication Anti-Spoofing Using Power-Spectra-Based Variance

(166) FIG. **12-1** illustrates an example scheme implemented by the ultrasonic sensor **216** to generate yet another single-channel or multi-channel feature for face authentication anti-spoofing. In this example, the ultrasonic sensor **216** uses spectral variance to evaluate an amount of variance observed over time within at least one receive channel **608**. Due to the planar structure of some presentation-attack instruments **404**, the spectral signature associated with the presentation-attack instrument **404** can have higher variance than distributed targets, such as the human face **408**. By analyzing spectral variance for one or more receive channels **608-1** to **608-M**, the ultrasonic sensor **216** can distinguish between the presentation-attack instrument **404** and the human face **408** for face authentication anti-spoofing.

(167) In the depicted configuration, the feature-extractor **704** includes a sliding window **1202** and a variance feature extractor **1204**. The spoofing predictor **706** can include the comparator **708**, the look-up table **710**, and/or the machine-learned module **712**. During operation, the pre-processor **702** (not shown) provides a power spectrum **722** associated with one of the receive channels **608** to the sliding window **1202**. To generate the power spectrum **722**, the pre-processor **702** can perform a one-dimensional Fourier transform operation on the range-profile data **716** across the range dimension. Alternatively, the pre-processor **702** can perform a two-dimensional Fourier transform operation on the range-slow time data **720**. Prior to generating the power spectrum **722**, the pre-processor **702** can filter the range-profile data **716** based on the measured location of the object. In this way, the pre-processor **702** can cause a later-determined single-channel feature **726** or multi-channel feature **728** to be associated with the object of interest.

(168) The sliding window **1202** divides the power spectrum **722** into individual subframes **1206** along a time dimension. These subframes **1206** have different time intervals that may or may not overlap. The variance feature extractor **1204** computes a standard deviation (or variance) of the amplitude information within one or more subframes and optionally within each subframe, to generate the variance feature **1208**. Example variance features **1208** are further described below with respect to FIGS. **12-2** and **12-3**.

(169) FIGS. **12-2** and **12-3** illustrate standard-deviation graphs **1210** to **1220** associated with face authentication anti-spoofing. The standard-deviation graphs **1210** and **1212** are respectively associated with a human face **408** and a human face **408** wearing a hat as an accessory **110**. The standard-deviation graph **1214** is associated with a plastic mask. The standard-deviation graphs **1216**, **1218**, and **1220** are respectively associated with a paper-based medium **116**, the device **120**, and a latex mask. Each of the standard-deviation graphs **1210** to **1220** depict variances **1222** and **1224** associated with receive channels **608-1** and **608-2** (e.g., associated with microphones **504-1**

and 504-2 of FIG. 5).

(170) To distinguish between the human face **408** (with or without the accessory **110**) and the various presentation-attack instruments **404**, the spoofing predictor **706** can determine whether the variances **1222** and **1224** are within a set of expected values. If the variances **1222** and **1224** are within the set of expected values, the spoofing predictor **706** determines that an object presented for face authentication **104** corresponds to the human face **408**. Alternatively, if the variances **1222** or **1224** have values outside of the set of expected values, the spoofing predictor **706** determines that the presented object corresponds to the presentation-attack instrument **404**. The set of expected values can generally include the values shown in the standard-deviation graphs **1210** and **1212**.

(171) Considering the standard-deviation graphs **1216** and **1218** associated with the paper-based medium **116** and the device **120**, the variances **1222** and **1224** are significantly larger than the variances **1222** and **1224** of the standard-deviation graphs **1210** and **1212**. For the standard-deviation graph **1220** associated with the latex mask, the variances **1222** and **1224** are significantly smaller than the variances **1222** and **1224** of the standard-deviation graphs **1210** and **1212**.

(172) It can be challenging to distinguish between the standard-deviation graphs **1210** and **1212** and the standard-deviation graph **1214** associated with the plastic mask using spectral variance. In this case, the ultrasonic sensor **216** can perform one or more of the other described techniques to distinguish between the human face **408** and the plastic mask.

(173) Example Methods

(174) FIGS. **13**, **14**, and **15** depict example methods **1300**, **1400**, and **1500** for face authentication anti-spoofing using ultrasound. Each method **1300**, **1400**, and **1500** is shown as a set of operations (or acts) performed and is not necessarily limited to the order or combinations in which the operations are shown herein. Further, any of one or more of the operations may be repeated, combined, reorganized, or linked to provide a wide array of additional and/or alternative methods. In portions of the following discussion, reference may be made to environments **100-1** to **100-5** of FIG. **1**, and entities detailed in FIGS. **2** to **4-1**, reference to which is made for example only. The techniques are not limited to performance by one entity or multiple entities operating on one user device **102**.

(175) At **1302** in FIG. **13**, an ultrasonic transmit signal is transmitted. For example, the ultrasonic sensor **216** transmits the ultrasonic transmit signal **410**, as shown in FIG. **4-1**. The ultrasonic transmit signal **410** includes frequencies in the range between approximately 20 kHz to 2 MHz, for example as described with reference to FIG. **2-2** above, and can represent a pulsed signal or a continuous signal. In some cases, the ultrasonic sensor **216** modulates a characteristic of the ultrasonic transmit signal **410**, including the phase and/or frequency. In some implementations, the ultrasonic sensor **216** transmits the ultrasonic transmit signal **410-1** or **410-S** responsive to the face-authentication system **202** receiving the request **302** or responsive to the ultrasonic sensor **216** receiving an alert from the sensor **218** (e.g., an alert from the inertial measurement unit **236**) indicating that the user device **102** is approximately stationary.

(176) The ultrasonic sensor **216** can use a dedicated transducer **222** to transmit the ultrasonic transmit signal **410**. In other implementations, the ultrasonic sensor **216** can use a shared speaker (e.g., speaker **502**) of the user device **102** to transmit the ultrasonic transmit signal **410**.

(177) At **1304**, an ultrasonic receive signal is received. The ultrasonic receive signal comprises a version of the ultrasonic transmit signal that is reflected by an object. For example, the ultrasonic sensor **216** receives the ultrasonic receive signal **412** using one of the microphones **504-1** or **504-2**. The ultrasonic receive signal **412** is a version of the ultrasonic transmit signal **410** (e.g., a delayed version of the ultrasonic transmit signal **410**), which is reflected by an object (e.g., the presentation-attack instrument **404** or the human face **408**). In some cases, the ultrasonic receive signal **412** has a different amplitude than the ultrasonic transmit signal **410** and/or is shifted in phase and/or frequency. In some implementations, the ultrasonic sensor **216** receives the ultrasonic receive signal **412** during at least a portion of time that the ultrasonic transmit signal **410** is transmitted.

(178) At **1306**, range-profile data is generated based on the ultrasonic receive signal. The range-profile data comprises amplitude and phase information associated with the ultrasonic receive signal. For example, the spoofing detector **230** of the ultrasonic sensor **216** generates the range-profile data **716**, which is a type of complex data **714**. The range-profile data **716** includes amplitude and phase information across a range dimension and a time dimension.

(179) At **1308**, a feature of the range-profile data within a region-of-interest that is associated with a location of the object is determined. For example, the ultrasonic sensor **216** determines a feature **724** (e.g., a single-channel feature **726**) within a region-of-interest, which is associated with a location of the object. Example single-channel features **726** can include the peak-amplitude feature **808**, the energy-distribution feature **810**, and/or the phase feature **812**, as shown in FIG. **8-1**. In general, the region-of-interest identifies a general spatial region that encompasses the location of the object. In some cases, the identified spatial region can surround at least a portion of the location of the object to account for a margin of error in determining the location of the object. An example spatial region can include the range **826** of the object shown in FIG. **8-2**. In some implementations, the ultrasonic sensor **216** can directly measure the location of the object by analyzing the range-profile data **716**. Additionally or alternatively, the ultrasonic sensor **216** can reference the sensor data **308** provided by the sensor **218** to determine the location of the object.

(180) At **1310**, a presentation attack is detected based on the feature. The presentation attack attempts to spoof a face-authentication system. The object is associated with the presentation attack. For example, the ultrasonic sensor **216** detects the presentation attack **114** (e.g., or determines the presence of the presentation-attack instrument **404**) based on the feature **724**, as described with respect to FIGS. **8-1** and **8-2**.

(181) The presentation attack **114** attempts to spoof the face-authentication system **202**. The object is associated with the presentation attack **114** and represents the presentation-attack instrument **404**. Example presentation-attack instruments **404** can include a photograph **118** of the user **106**, a digital picture **124** of the user **106**, or a mask **126** that replicates one or more features of the user **106**.

(182) At **1312**, the face-authentication system is prevented from authenticating the presentation attack. For example, the ultrasonic sensor **216** prevents the face-authentication system **202** from authenticating the presentation attack **114**. In particular, the ultrasonic sensor **216** provides the spoofing indicator **304** to the face recognizer **214**, which causes the face recognizer **214** to report **306** a face-authentication failure.

(183) In a variant of the method **1300**, block **1310** may instead determine based on the feature **724** that there is no presentation attack **114** (e.g., that the object that reflected the ultrasonic transmit signal **410** to generate the ultrasonic receive signal **412** is a human face). In this case, at **1312**, the face-authentication system **202** can authenticate the human face **408** to allow the user **106** access.

(184) By analyzing the range-profile data, the ultrasonic sensor **216** can perform face authentication anti-spoofing using as few as one receive channel **608**. This can be useful for space-constrained devices in which the ultrasonic sensor **216** is implemented with a single receive channel **608**. If additional receive channels **608** are available, the ultrasonic sensor **216** can evaluate the features **724** associated with one or more receive channels and optionally with each receive channel **608** to improve its ability to correctly detect the presentation attack **114**.

(185) At **1402** in FIG. **14**, an ultrasonic transmit signal is transmitted using an ultrasonic sensor. For example, the ultrasonic sensor **216** transmits the ultrasonic transmit signal **410**, as described in FIG. **13** with respect to **1302**.

(186) At **1404**, at least two ultrasonic receive signals are received using at least two transducers of an ultrasonic sensor. The at least two ultrasonic receive signals comprise respective versions of the ultrasonic transmit signal that is reflected by an object. For example, the ultrasonic sensor **216** receives at least two of the ultrasonic receive signals **412-1** to **412-M** using at least two of the microphones **504-1** to **504-M**. The ultrasonic receive signal **412-1** to **412-M** are respective versions

of the ultrasonic transmit signal **410**, which is reflected by an object (e.g., the presentation-attack instrument **404** or the human face **408**). The ultrasonic receive signals **412-1** to **412-M** can have different amplitudes than the ultrasonic transmit signal **410** and/or can be shifted in phase and/or frequency. In some implementations, the ultrasonic sensor **216** receives the ultrasonic receive signals **412-1** to **412-M** during at least a portion of time that the ultrasonic transmit signal **410** is transmitted.

(187) At **1406**, an interferogram is generated based on the at least two ultrasonic receive signals. The interferogram comprises coherence information and phase information. For example, the spoofing detector **230** of the ultrasonic sensor **216** generates the interferogram **718** based on the at least two ultrasonic receive signals **412-1** to **412-M**, as shown in FIG. **9-1**. In particular, the pre-processor **702** of the spoofing detector **230** generates the interferogram **718** by combining information from range-profile data **716** associated with two of the receive channels **608-1** to **608-M**. The interferogram **718** includes coherence information **936** and phase information **934**, as shown in FIG. **9-2**.

(188) At **1408**, a coherence feature is identified based on the coherence information of the interferogram. The coherence feature represents an amount of coherence within a region-of-interest. For example, the spoofing detector **230** of the ultrasonic sensor **216** identifies (e.g., extracts or generates) the coherence feature **914**, which represents an amount of coherence within a region-of-interest **938**. The coherence feature **914** can indicate a significantly large amount of coherence if the object is associated with a presentation-attack instrument **404** (as shown in interferograms **930** or **932**) or a substantially small amount of coherence if the object is associated with the human face **408** (as shown in interferogram **928**). An example coherence feature **914** can represent an average coherence within the region-of-interest **938**. Another example coherence feature **914** represents an amount of coherence variance within the region-of-interest **938**.

(189) At **1410**, a presentation attack is detected based on the coherence feature. The presentation attack attempts to spoof a face-authentication system. The object is associated with the presentation attack. For example, the ultrasonic sensor **216** detects the presentation attack **114** based on the coherence feature **914**, as shown in FIG. **9-1**. The presentation attack **114** attempts to spoof the face-authentication system **202**. The object is associated with the presentation attack **114**.

(190) At **1412**, the face-authentication system is prevented from authenticating the presentation attack **114**. For example, the ultrasonic sensor **216** prevents the face-authentication system **202** from authenticating the presentation attack **114**. In particular, the ultrasonic sensor **216** provides the spoofing indicator **304** to the face recognizer **214**, which causes the face recognizer **214** to report **306** a face-authentication failure.

(191) In a variant of the method **1400**, block **1410** may instead determine based on the coherence feature **914** that there is no presentation attack **114** (e.g., that the object that reflected the ultrasonic transmit signal **410** to generate the ultrasonic receive signal **412** is a human face). In this case, at **1412**, the face-authentication system **202** can authenticate the human face **408** to allow the user **106** access.

(192) At **1502** in FIG. **15**, an ultrasonic transmit signal is transmitted. For example, the ultrasonic sensor **216** transmits the ultrasonic transmit signal **410**, as described in FIG. **13** with respect to **1302**.

(193) At **1504**, at least two ultrasonic receive signals are received using at least two transducers of an ultrasonic sensor. The at least two ultrasonic receive signals comprise respective versions of the ultrasonic transmit signal that is reflected by an object. For example, the ultrasonic sensor **216** receives at least two of the ultrasonic receive signals **412-1** to **412-M** using at least two of the microphones **504-1** to **504-M**, as described in FIG. **14** with respect to **1404**.

(194) At **1506**, power spectra are generated based on the at least two ultrasonic receive signals. The power spectra represent power of the at least two ultrasonic receive signals over a set of frequencies and a time interval. For example, the spoofing detector **230** of the ultrasonic sensor **216**

generates the power spectra **722**, as shown in FIG. **12-1**. The power spectra **722** represents power of the at least two ultrasonic receive signals over a set of frequencies and a time interval.

(195) At **1508**, variances of the power within the power spectra are determined over time. The variances are respectively associated with the at least two ultrasonic receive signals. For example, the spoofing detector **230** generates a variance feature **1208**, which represents a standard deviation of the power within the power spectra over time. Example standard deviation graphs are shown in FIGS. **12-2** and **12-3** for different types of objects.

(196) At **1510**, a presentation attack is detected based on the variances. The presentation attack attempts to spoof a face-authentication system. The object is associated with the presentation attack. For example, the spoofing detector **230** detects the presentation attack **114** based on the variance feature **1208**, as shown in FIG. **12-1**. The presentation attack **114** attempts to spoof the face-authentication system **202**. The object is associated with the presentation attack **114**.

(197) At **1512**, the face-authentication system is prevented from authenticating the presentation attack. For example, the ultrasonic sensor **216** prevents the face-authentication system **202** from authenticating the presentation attack **114**. In particular, the ultrasonic sensor **216** provides the spoofing indicator **304** to the face recognizer **214**, which causes the face recognizer **214** to report **306** a face-authentication failure.

(198) In a variant of the method **1500**, block **1510** may instead determine based on the variance feature **1208** that there is no presentation attack **114** (e.g., that the object that reflected the ultrasonic transmit signal **410** to generate the ultrasonic receive signal **412** is a human face). In this case, at **1512**, the face-authentication system **202** can authenticate the human face **408** to allow the user **106** access.

(199) The operations of FIGS. **13** to **15** can be combined in various ways. In general, the ultrasonic sensor **216** can perform face authentication anti-spoofing using any combination of single-channel features **726** or multi-channel features **728**. Although employing additional techniques for face authentication anti-spoofing can use additional computational resources, it can increase an accuracy of the ultrasonic sensor **216** correctly distinguishing between the human face **408** and the presentation-attack instrument **404**. Furthermore, performing multiple techniques can better enable the ultrasonic sensor **216** to distinguish between the human face **408** wearing an accessory **110** and the presentation attack **114**.

(200) Example Computing System

(201) FIG. **16** illustrates various components of an example computing system **1600** that can be implemented as any type of client, server, and/or user device **102** as described with reference to the previous FIG. **2-1** to perform face authentication anti-spoofing using an ultrasonic sensor **216**.

(202) The computing system **1600** includes communication devices **1602** that enable wired and/or wireless communication of device data **1604** (e.g., received data, data that is being received, data scheduled for broadcast, or data packets of the data). The communication devices **1602** or the computing system **1600** can include one or more ultrasonic sensors **216** and one or more sensor **218**. The device data **1604** or other device content can include configuration settings of the device, media content stored on the device, and/or information associated with a user **106** of the device. Media content stored on the computing system **1600** can include any type of audio, video, and/or image data. The computing system **1600** includes one or more data inputs **1606** via which any type of data, media content, and/or inputs can be received, including human utterances, inputs from the ultrasonic sensor **216**, user-selectable inputs (explicit or implicit), messages, music, television media content, recorded video content, and any other type of audio, video, and/or image data received from any content and/or data source.

(203) The computing system **1600** also includes communication interfaces **1608**, which can be implemented as any one or more of a serial and/or parallel interface, a wireless interface, any type of network interface, a modem, and as any other type of communication interface. The communication interfaces **1608** provide a connection and/or communication links between the

computing system **1600** and a communication network by which other electronic, computing, and communication devices communicate data with the computing system **1600**.

(204) The computing system **1600** includes one or more processors **1610** (e.g., any of microprocessors, controllers, and the like), which process various computer-executable instructions to control the operation of the computing system **1600** and to enable techniques for, or in which can be embodied, face-authentication anti-spoofing using ultrasound. Alternatively or in addition, the computing system **1600** can be implemented with any one or combination of hardware, firmware, or fixed logic circuitry that is implemented in connection with processing and control circuits, which are generally identified at **1612**. Although not shown, the computing system **1600** can include a system bus or data transfer system that couples the various components within the device. A system bus can include any one or combination of different bus structures, including a memory bus or memory controller, a peripheral bus, a universal serial bus, and/or a processor or local bus that utilizes any of a variety of bus architectures.

(205) The computing system **1600** also includes a computer-readable medium **1614**, including one or more memory devices that enable persistent and/or non-transitory data storage (i.e., in contrast to mere signal transmission), examples of which include random access memory (RAM), non-volatile memory (e.g., any one or more of a read-only memory (ROM), flash memory, EPROM, EEPROM, etc.), and a disk storage device. The disk storage device may be implemented as any type of magnetic or optical storage device, including a hard disk drive, a recordable and/or rewriteable compact disc (CD), any type of a digital versatile disc (DVD), and the like. The computing system **1600** can also include a mass storage media device (storage media) **1616**.

(206) The computer-readable medium **1614** provides data storage mechanisms to store the device data **1604**, as well as various device applications **1618** and any other types of information and/or data related to operational aspects of the computing system **1600**. For example, an operating system **1620** can be maintained as a computer application with the computer-readable medium **1614** and executed on the processors **1610**. The device applications **1618** may include a device manager, including any form of a control application, software application, signal-processing and control module, code that is native to a particular device, a hardware abstraction layer for a particular device, and so on. Using the ultrasonic sensor **216**, the computing system **1600** can perform face authentication anti-spoofing using interferometry-based coherence.

CONCLUSION

(207) Although techniques using, and apparatuses including, an ultrasonic sensor to perform face authentication anti-spoofing using interferometry-based coherence have been described in language specific to features and/or methods, it is to be understood that the subject of the appended claims is not necessarily limited to the specific features or methods described. Rather, the specific features and methods are disclosed as example implementations of face authentication anti-spoofing using interferometry-based coherence.

(208) Some Examples are described below.

(209) Example 1: A method performed by an ultrasonic sensor, the method comprising:

(210) transmitting an ultrasonic transmit signal;

(211) receiving an ultrasonic receive signal, the ultrasonic receive signal comprising a version of the ultrasonic transmit signal that is reflected by an object;

(212) generating range-profile data based on the ultrasonic receive signal, the range-profile data comprising amplitude and phase information associated with the ultrasonic receive signal;

(213) determining a feature of the range-profile data within a region-of-interest that is associated with a location of the object;

(214) detecting a presentation attack based on the feature, the presentation attack attempting to spoof a face-authentication system, the object associated with the presentation attack; and

(215) preventing the face-authentication system from authenticating the presentation attack.

(216) Example 2: The method of example 1, wherein:

(217) the determining of the feature comprises detecting a peak amplitude within the region-of-interest of the range-profile data; and

(218) the detecting the presentation attack comprises: comparing the peak amplitude to a threshold; and responsive to the peak amplitude being above the threshold, detecting the presentation attack.

(219) Example 3: The method of example 2, further comprising:

(220) transmitting another ultrasonic transmit signal;

(221) receiving another ultrasonic receive signal, the other ultrasonic receive signal comprising a version of the other ultrasonic transmit signal that is reflected by a human face;

(222) generating other range-profile data based on the other ultrasonic receive signal, the other range-profile data comprising amplitude and phase information associated with the other ultrasonic receive signal;

(223) detecting another peak amplitude within another region-of-interest of the other range-profile data, the other region-of-interest associated with a location of the human face;

(224) comparing the other peak amplitude to the threshold; and

(225) responsive to the other peak amplitude being less than the threshold, enabling the face-authentication system to authenticate the human face.

(226) Example 4: The method of any preceding example, wherein:

(227) the determining of the feature comprises determining a distribution of energy over the region-of-interest; and

(228) the detecting the presentation attack comprises detecting the presentation attack responsive to the distribution of energy being greater than another threshold.

(229) Example 5: The method of any preceding example, further comprising:

(230) determining the location of the object,

(231) wherein the region-of-interest includes the location of the object.

(232) Example 6: The method of example 5, further comprising:

(233) accepting sensor data from a sensor,

(234) wherein the determining the location of the object comprises determining the location of the object based on the sensor data.

(235) Example 7: The method of example 6, wherein the sensor comprises:

(236) a phase-difference sensor of a camera; or

(237) a radio-frequency sensor.

(238) Example 8: The method of any one of examples 5 to 7, further comprising:

(239) prior to determining the feature, filtering the range-profile data to extract the amplitude and phase information associated with the region-of-interest.

(240) Example 9: The method of any preceding example, wherein the transmitting of the ultrasonic transmit signal comprises generating the ultrasonic transmit signal having a bandwidth that results in a range resolution of the range-profile data being less than approximately 7 centimeters.

(241) Example 10: The method of any preceding example, further comprising:

(242) determining a distance to the object; and

(243) prior to detecting the presentation attack, normalizing the amplitude information within the range-profile data based on the distance to the object.

(244) Example 11: The method of any one of examples 1 to 9, further comprising:

(245) prior to detecting the presentation attack, normalizing the amplitude information within the range-profile data based on a cross-coupling factor or a constant-false-alarm-rate signal-to-noise ratio.

(246) Example 12: The method of any preceding example, further comprising:

(247) accepting motion data from an inertial measurement unit, the motion data representing motion of the ultrasonic sensor; and

(248) compensating for the motion of the ultrasonic sensor by modifying the range-profile data based on the motion data.

(249) Example 13: The method of any one of examples 1 to 11, further comprising:

(250) accepting motion data from an inertial measurement unit; and

(251) determining that the ultrasonic sensor is substantially stationary based on the motion data,

(252) wherein the transmitting the ultrasonic transmit signal comprises transmitting the ultrasonic transmit signal responsive to determining that the ultrasonic sensor is substantially stationary.

(253) Example 14: An apparatus comprising an ultrasonic sensor configured to perform any one of the methods of examples 1 to 13.

(254) Example 15: A computer-readable medium comprising instructions which, when executed by a processor, cause an ultrasonic sensor to perform any one of the methods of examples 1 to 13.

(255) Example 16: A method performed by an ultrasonic sensor, the method comprising:

(256) transmitting an ultrasonic transmit signal;

(257) receiving at least two ultrasonic receive signals using at least two transducers of the ultrasonic sensor, the at least two ultrasonic receive signals comprising respective versions of the ultrasonic transmit signal that is reflected by an object;

(258) generating an interferogram based on the at least two ultrasonic receive signals, the interferogram comprising coherence information and phase information;

(259) identifying a coherence feature based on the coherence information of the interferogram, the coherence feature representing an amount of coherence within a region-of-interest;

(260) detecting a presentation attack based on the coherence feature, the presentation attack attempting to spoof a face-authentication system, the object associated with the presentation attack; and

(261) preventing the face-authentication system from authenticating the presentation attack.

(262) Example 17: The method of example 16, wherein the detecting the presentation attack comprises, responsive to the amount of coherence being greater than a coherence threshold, detecting the presentation attack.

(263) Example 18: The method of example 17, wherein the detecting the presentation attack comprises:

(264) transmitting another ultrasonic transmit signal;

(265) receiving at least two other ultrasonic receive signals, the at least two other ultrasonic receive signals comprising versions of the other ultrasonic transmit signal that is reflected by a human face;

(266) generating another interferogram based on the at least two other ultrasonic receive signals, the other interferogram comprising coherence and phase information;

(267) identifying another coherence feature based on the coherence information of the other interferogram, the other coherence feature representing another amount of coherence within another region-of-interest; and

(268) responsive to the other amount of coherence being less than the coherence threshold, enabling the face-authentication system to authenticate the human face.

(269) Example 19: The method of one of examples 16 to 18, wherein the detecting the presentation attack comprises:

(270) identifying a phase feature based on the phase information of the interferogram, the phase feature representing an amount of phase variation within the region-of-interest; and

(271) responsive to the amount of phase variation being less than a phase threshold, detecting the presentation attack.

(272) Example 20: The method of any one of examples 16 to 19, further comprising:

(273) generating range-profile data based on the at least two ultrasonic receive signals; and

(274) performing co-registration on the range-profile data to generate co-registered range-profile data,

(275) wherein the generating the interferogram comprises generating the interferogram based on the co-registered range-profile data.

(276) Example 21: The method of example 20, wherein:

(277) the range-profile data comprises: first range-profile data associated with a first ultrasonic receive signal of the at least two ultrasonic receive signals; and second range-profile data associated with a second ultrasonic receive signal of the at least two ultrasonic receive signals; and

(278) the performing of the co-registration aligns respective responses of the object within the first range-profile data and the second range-profile data along a range dimension.

(279) Example 22: The method of example 21, wherein the performing of the co-registration comprises:

(280) filtering the first range-profile data through a first low-pass filter; and

(281) filtering the second range-profile data through a second low-pass filter.

(282) Example 23: The method of example 21, wherein the performing of the co-registration comprises:

(283) detecting a first peak amplitude within the first range-profile data;

(284) determining a first range bin associated with the first peak amplitude;

(285) detecting a second peak amplitude within the second range-profile data;

(286) determining a second range bin associated with the second peak amplitude;

(287) determining an offset based on a difference between the first range bin and the second range bin; and

(288) shifting the second range-profile data in range based on the offset to generate shifted second range-profile data.

(289) Example 24: The method of example 23, wherein the performing of the co-registration comprises:

(290) determining another offset between the first range-profile data and the shifted second range-profile data; and

(291) interpolating the shifted second range-profile data based on the other offset to generate interpolated and shifted second range-profile data.

(292) Example 25: The method of example 21, wherein the performing of the co-registration comprises:

(293) resampling the second range-profile data to project the second range-profile data to a common grid associated with the first range-profile data.

(294) Example 26: The method of any one of examples 20 to 25, wherein the detecting the presentation attack comprises:

(295) generating power distribution histograms based on the range-profile data, the power distribution histograms respectively associated with the at least two ultrasonic receive signals; and

(296) detecting the presentation attack based on skewness of the power-distribution histograms.

(297) Example 27: The method of any one of examples 20 to 26, wherein the detecting the presentation attack comprises:

(298) generating constant-false-alarm-rate signal-to-noise ratio distribution histograms based on the range-profile data, the constant-false-alarm-rate signal-to-noise ratio distribution histograms respectively associated with the at least two ultrasonic receive signals; and

(299) detecting the presentation attack based on skewness of the constant-false-alarm-rate signal-to-noise ratio distribution histograms.

(300) Example 28: The method of example 27, wherein the shapes of the constant-false-alarm-rate signal-to-noise ratio distribution histograms exhibit one or more of the following:

(301) a non-Gaussian distribution; or

(302) a distribution with a tail.

(303) Example 29: An apparatus comprising an ultrasonic sensor configured to perform any one of the methods of examples 16 to 28.

(304) Example 30: The apparatus of example 29, wherein:

(305) the apparatus comprises a smartphone;

(306) the ultrasonic sensor is integrated within the smartphone; and

(307) the at least two transducers of the ultrasonic sensor comprise: a first microphone of the smartphone; and a second microphone of the smartphone.

(308) Example 31: The apparatus of example 30, wherein the first microphone and the second microphone are positioned on opposite ends of the smartphone.

(309) Example 32: The apparatus of any one of examples 29 to 31, wherein a distance between the at least two transducers is greater than a wavelength associated with the ultrasonic transmit signal.

(310) Example 33: A computer-readable medium comprising instructions which, when executed by a processor, cause an ultrasonic sensor to perform any one of the methods of examples 16 to 28.

(311) Example 34: A method performed by an ultrasonic sensor, the method comprising:

(312) transmitting an ultrasonic transmit signal;

(313) receiving at least two ultrasonic receive signals using at least two transducers of the ultrasonic sensor, the at least two ultrasonic receive signals comprising respective versions of the ultrasonic transmit signal that is reflected by an object;

(314) generating power spectra based on the at least two ultrasonic receive signals, the power spectra representing power of the at least two ultrasonic receive signals over a set of frequencies and a time interval;

(315) determining variances of the power within the power spectra over time, the variances respectively associated with the at least two ultrasonic receive signals;

(316) detecting a presentation attack based on the variances, the presentation attack attempting to spoof a face-authentication system, the object associated with the presentation attack; and

(317) preventing the face-authentication system from authenticating the presentation attack.

(318) Example 35: The method of example 34, wherein the generating the power spectra comprises:

(319) generating complex data based on the at least two ultrasonic receive signals; and

(320) performing a Fourier transform to generate the power spectra.

(321) Example 36: The method of example 34 or 35, wherein the complex data comprises:

(322) range-profile data;

(323) range-slow time data; or

(324) an interferogram.

(325) Example 37: The method of any one of examples 34 to 36, wherein the detecting the presentation attack comprises:

(326) determining whether the variances are within a set of values; and

(327) detecting the presentation attack responsive to at least one of the variances being outside of the set of values.

(328) Example 38: The method of example 37, wherein the detecting the presentation attack comprises:

(329) transmitting another ultrasonic transmit signal;

(330) receiving at least two other ultrasonic receive signals, the at least two other ultrasonic receive signals comprising versions of the other ultrasonic transmit signal that is reflected by a human face;

(331) generating other power spectra based on the at least two other ultrasonic receive signals, the other power spectra representing power of the at least two other ultrasonic receive signals over the set of frequencies and another time interval;

(332) determining other variances of the power within the other power spectra over time, the other variances respectively associated with the at least two ultrasonic receive signals; and

(333) responsive to the other variances being within the set of values, enabling the face-authentication system to authenticate the human face.

(334) Example 39: The method of example 38, wherein the human face is wearing an accessory.

(335) Example 40: The method of any one of examples 34 to 39, wherein:

(336) the power spectra comprises: a first power spectrum associated with a first ultrasonic receive signal of the at least two ultrasonic receive signals; and a second power spectrum associated with a

second ultrasonic receive signal of the at least two ultrasonic receive signals; and

(337) the determining the variances comprises: generating, using a sliding window, a first set of subframes of the first power spectrum, subframes of the first set of subframes associated with different portions of the time interval; computing a first standard deviation across the first set of subframes; generating, using the sliding window, a second set of subframes of the second power spectrum, subframes of the second set of subframes associated with the different portions of the time interval; and computing a second standard deviation across the second set of subframes.

(338) Example 41: The method of any one of examples 1 to 13, 16 to 28, or 34 to 40, wherein the presentation attack comprises:

(339) an unauthorized actor presenting a photograph of an authorized user;

(340) the unauthorized actor presenting a device that displays a digital picture of the authorized user; or

(341) the unauthorized actor wearing a mask that represents the authorized user.

(342) Example 42: An apparatus comprising an ultrasonic sensor configured to perform any one of the methods of examples 34 to 41.

(343) Example 43: The apparatus of example 42, wherein:

(344) the apparatus comprises a smartphone;

(345) the ultrasonic sensor is integrated within the smartphone; and

(346) the at least two transducers of the ultrasonic sensor comprise: a first microphone of the smartphone; and a second microphone of the smartphone.

(347) Example 44: The apparatus of example 43, wherein the first microphone and the second microphone are positioned on opposite ends of the smartphone.

(348) Example 45: The apparatus of any one of examples 42 to 44, wherein a distance between the at least two transducers is greater than a wavelength associated with the ultrasonic transmit signal.

(349) Example 46: A computer-readable medium comprising instructions which, when executed by a processor, cause an ultrasonic sensor to perform any one of the methods of examples 34 to 41.

Claims

1. A method performed by an ultrasonic sensor, the method comprising: transmitting an ultrasonic transmit signal; receiving at least two ultrasonic receive signals using at least two transducers of the ultrasonic sensor, the at least two ultrasonic receive signals comprising respective versions of the ultrasonic transmit signal that is reflected by an object; generating an interferogram based on the at least two ultrasonic receive signals, the interferogram comprising coherence information and phase information; identifying a coherence feature based on the coherence information of the interferogram, the coherence feature representing an amount of coherence within a region-of-interest; detecting a presentation attack based on a comparison of the coherence feature to a coherence threshold, the presentation attack attempting to spoof a face-authentication system, the object associated with the presentation attack; and preventing the face-authentication system from authenticating the presentation attack.
2. The method of claim 1, wherein the detecting the presentation attack comprises detecting the presentation attack based on the amount of coherence being greater than the coherence threshold.
3. The method of claim 2, wherein the detecting the presentation attack comprises: transmitting another ultrasonic transmit signal; receiving at least two other ultrasonic receive signals, the at least two other ultrasonic receive signals comprising versions of the other ultrasonic transmit signal that is reflected by a human face; generating another interferogram based on the at least two other ultrasonic receive signals, the other interferogram comprising coherence and phase information; identifying another coherence feature based on the coherence information of the other interferogram, the other coherence feature representing another amount of coherence within another region-of-interest; and responsive to the other amount of coherence being less than the

coherence threshold, enabling the face-authentication system to authenticate the human face.

4. The method of claim 1, wherein the detecting the presentation attack comprises: identifying a phase feature based on the phase information of the interferogram, the phase feature representing an amount of phase variation within the region-of-interest; and responsive to the amount of phase variation being less than a phase threshold, detecting the presentation attack.

5. The method of claim 1, further comprising: generating range-profile data based on the at least two ultrasonic receive signals; and performing co-registration on the range-profile data to generate co-registered range-profile data, wherein the generating the interferogram comprises generating the interferogram based on the co-registered range-profile data.

6. The method of claim 5, wherein: the range-profile data comprises: first range-profile data associated with a first ultrasonic receive signal of the at least two ultrasonic receive signals; and second range-profile data associated with a second ultrasonic receive signal of the at least two ultrasonic receive signals; and the performing of the co-registration aligns respective responses of the object within the first range-profile data and the second range-profile data along a range dimension.

7. The method of claim 6, wherein the performing of the co-registration comprises: filtering the first range-profile data through a first low-pass filter; and filtering the second range-profile data through a second low-pass filter.

8. The method of claim 6, wherein the performing of the co-registration comprises: detecting a first peak amplitude within the first range-profile data; determining a first range bin associated with the first peak amplitude; detecting a second peak amplitude within the second range-profile data; determining a second range bin associated with the second peak amplitude; determining an offset based on a difference between the first range bin and the second range bin; and shifting the second range-profile data in range based on the offset to generate shifted second range-profile data.

9. The method of claim 8, wherein the performing of the co-registration comprises: determining another offset between the first range-profile data and the shifted second range-profile data; and interpolating the shifted second range-profile data based on the other offset to generate interpolated and shifted second range-profile data.

10. The method of claim 6, wherein the performing of the co-registration comprises: resampling the second range-profile data to project the second range-profile data to a common grid associated with the first range-profile data.

11. The method of claim 5, wherein the detecting the presentation attack comprises: generating power distribution histograms based on the range-profile data, the power distribution histograms respectively associated with the at least two ultrasonic receive signals; and detecting the presentation attack based on skewness of the power distribution histograms.

12. The method of claim 5, wherein the detecting the presentation attack comprises: generating constant-false-alarm-rate signal-to-noise ratio distribution histograms based on the range-profile data, the constant-false-alarm-rate signal-to-noise ratio distribution histograms respectively associated with the at least two ultrasonic receive signals; and detecting the presentation attack based on skewness of the constant-false-alarm-rate signal-to-noise ratio distribution histograms.

13. The method of claim 12, wherein shapes of the constant-false-alarm-rate signal-to-noise ratio distribution histograms exhibit one or more of the following: a non-Gaussian distribution; or a distribution with a tail.

14. An apparatus comprising: an ultrasonic sensor comprising: multiple transducers configured to: transmit an ultrasonic transmit signal using at least one transducer of the multiple transducers; and receive at least two ultrasonic receive signals using at least two transducers of the multiple transducers, the at least two ultrasonic receive signals comprising respective versions of the ultrasonic transmit signal that is reflected by an object; and at least one processor configured to: generate an interferogram based on the at least two ultrasonic receive signals, the interferogram comprising coherence information and phase information; identify a coherence feature based on the

coherence information of the interferogram, the coherence feature representing an amount of coherence within a region-of-interest; detect a presentation attack based on a comparison of the coherence feature to a coherence threshold, the presentation attack attempting to spoof a face-authentication system, the object associated with the presentation attack; and prevent the face-authentication system from authenticating the presentation attack.

15. The apparatus of claim 14, wherein: the apparatus comprises a smartphone; the ultrasonic sensor is integrated within the smartphone; and the at least two transducers of the ultrasonic sensor comprise: a first microphone of the smartphone; and a second microphone of the smartphone.

16. The apparatus of claim 15, wherein the first microphone and the second microphone are positioned on opposite ends of the smartphone.

17. The apparatus of claim 14, wherein a distance between the at least two transducers is greater than a wavelength associated with the ultrasonic transmit signal.

18. The apparatus of claim 14, wherein the ultrasonic sensor is configured to detect the presentation attack based on the amount of coherence being greater than the coherence threshold.

19. The apparatus of claim 14, wherein the ultrasonic sensor is configured to: identify a phase feature based on the phase information of the interferogram, the phase feature representing an amount of phase variation within the region-of-interest; and responsive to the amount of phase variation being less than a phase threshold, detect the presentation attack.

20. A non-transitory computer-readable medium comprising instructions which, when executed by a processor, cause an ultrasonic sensor to: generate an interferogram based on at least two ultrasonic receive signals, the interferogram comprising coherence information and phase information, the at least two ultrasonic receive signals comprising respective versions of an ultrasonic transmit signal that is reflected by an object and received by at least two transducers of the ultrasonic sensor; identify a coherence feature based on the coherence information of the interferogram, the coherence feature representing an amount of coherence within a region-of-interest; detect a presentation attack based on a comparison of the coherence feature to a coherence threshold, the presentation attack attempting to spoof a face-authentication system, the object associated with the presentation attack; and prevent the face-authentication system from authenticating the presentation attack.
